

Profiles of Debt: Revolving Behavior and Financial Well-Being

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1 Introduction and Related Literature

1.1 Introduction

By the second quarter of 2024, total credit card debt in Canada had exceeded CAD 122 billion, the highest since 2007 (Equifax Canada, 2024). This growing debt burden is not unique to Canada, but is also echoed across other major advanced economies. In the United States, for instance, total credit card debt surpassed USD 1 trillion for the first time in 2023 (Federal Reserve Bank of New York, 2026), with similar patterns observed in the United Kingdom (Bank of England, 2026). Against the background of rising living costs and broader economic uncertainty, this mounting debt pressure has intensified concerns about consumer well-being.

The prevailing narrative often portrays cardholders who carry revolving balances (i.e., *Revolvers*) as financially constrained consumers who rely on high-interest debt to make ends meet (CBC News, 2022) and lack sufficient savings to absorb unexpected economic shocks (Fulford and Schuh, 2024). As a result, public discourse frequently treats revolving balances as a primary indicator of financial distress (Bank of Canada, 2023b; Zhao, Witts, and Xiao, 2026).

Yet, little is known about the financial health of *Revolvers* relative to their non-revolving counterparts (i.e., *Convenience users*), despite its importance for effective policy interventions. This gap largely reflects limited data linking credit card borrowing behavior to household financial standing. Consider the policies mandating higher minimum monthly payments (key2019; Allen, Boutros, and Guttman-Kenney, 2024), for example. If *Revolvers* are indeed in more fragile financial states, higher minimum payment thresholds can reduce the debt burden by curbing long-term debt accumulation (Allen, Boutros, and Guttman-Kenney, 2024). By contrast, should consumers carry debt as a way to smooth their consumption over time, mandating higher minimum payments could unnecessarily restrict their available cash flow and ultimately reduce overall welfare.¹

This paper fills an important evidence gap regarding *Revolvers*' financial standing by leveraging rich microdata on various measures of credit card holders' financial well-being. In doing so, it offers a data-driven reevaluation of the prevailing narratives about the economic fragility of revolving

¹Allen, Boutros, and Guttman-Kenney, 2024 also finds that these reductions may come at the cost of greater delinquency and more restricted credit access.

credit users. Beyond documenting the financial standing of *Revolvers*, the paper also explores potential mechanisms underlying consumers' borrowing decisions and examines the socio-demographic determinants of revolving behavior.

The main data source for this paper is the Bank of Canada's Digital Wallet and Payment Trends survey, an annual cross-sectional survey administered by IPSOS. The dataset contains detailed individual-level information on spending, revolving status, demographic characteristics, and household financial conditions. This combination of credit card usage information and broader household balance-sheet measures makes the survey particularly useful for assessing whether revolving balances are systematically associated with financial distress.

First, the data provide limited support for the common interpretation of revolving balances as an early indicator of financial distress (Elul et al., 2010; Grodzicki and Koulayev, 2021; Zhao, Witts, and Xiao, 2026). Revolving and non-revolving cardholders have nearly identical distributions of financial asset holdings and only modest differences in liabilities. Consistent with these balance-sheet similarities, both groups also report comparable levels of financial comfort, with approximately 80% of cardholders in each group indicating that they are comfortable with their finances. This similarity persists after flexibly controlling for income, assets, liabilities, and other observable characteristics.

Second, although *Revolvers* report lower income on average than *Convenience users*, this difference largely reflects the correlation between income and other characteristics, rather than an independent relationship between the former and revolving behaviors. Results highlight that income is a weak predictor of revolving behavior along both the extensive (i.e., the propensity to revolve) and intensive (i.e., the amount borrowed conditional on revolving) margins. Conditional on asset and liability levels, a CAD 10,000 increase in pretax income reduces the likelihood of revolving by less than one percentage point (pp) relative to the 21% sample average, and it yields no statistically significant effect on the size of the balance carried.

By contrast, variables related to household asset composition are more strongly associated with revolving behavior. Investment account ownership predicts a 10 pp lower probability of revolving, whereas being a mortgagor predicts a 7 pp higher likelihood of revolving. One possible interpretation is that these variables capture differences in household liquidity. Since mortgage payments draw directly on disposable income, they inherently tighten

short-term cash flow constraints, holding other factors constant. Investment accounts, by contrast, may reflect greater financial flexibility, especially when they consist of relatively liquid assets such as stocks and mutual funds. Under this interpretation, mortgage and investment account ownership proxy for opposite liquidity conditions among cardholders. This perspective is consistent with the similarly sized but opposite effects of these variables on propensity to revolve, and echoes previous work emphasizing the role of liquidity in predicting borrowing behavior (Zhou, 2022; Islam and Picault, 2025). However, because the data do not contain direct measures of respondents’ liquidity constraints, these interpretations should be approached with caution.

Third, this paper tests whether *Revolvers* are indeed vulnerable to steep consumption decline in retirement (Angeletos et al., 2001). This concern rests on the standard prediction that patient *Convenience users* smooth their consumption through savings, while impatient *Revolvers* front-load consumption and exhibit heavily hump-shaped profiles (Fulford and Schuh, 2024). Because no existing long panel datasets offer sufficient detail on both consumption and outstanding balances, I empirically test these dynamics by constructing a synthetic cohort (Deaton, 1985; Gourinchas and Parker, 2002; Dang and Lanjouw, 2023). Specifically, I group respondents by age and revolving status and use older individuals observed in the same survey periods to approximate the future outcomes of otherwise comparable younger individuals. The approach yields a plausible hump-shaped income profiles for both revolving and non-revolving cardholders, thereby offering empirical support for the underlying framework.

The constructed consumption profiles contradict the standard prediction that *Revolvers* exhibit a more pronounced hump-shaped consumption path. Instead, *Revolvers* consume more early in life but experience significantly slower consumption growth during their prime working years. By contrast, *Convenience users* display a more conventional hump-shaped trajectory, with stronger consumption growth during working age. These patterns are consistent with a life-cycle model in which revolving and non-revolving consumers have similar lifetime resources, but the former are relatively more impatient.

At first glance, the finding that *Revolvers* appear to face tighter liquidity constraints while reporting similar levels of financial comfort may seem contradictory. To reconcile this pattern with the broader evidence on asset holdings, borrowing, and consumption, Section 6 puts forth two possible explanations.

The first explanation views revolving credit as an ex-post financial stop-

gap that allows liquidity-constrained consumers to meet other financial obligations without sharply reducing consumption (Telyukova, 2013; Greene and Stavins, 2023).

Alternatively, the second explanation considers the tighter liquidity constraints among *Revolvers* as ex-ante commitment devices, with credit card debt representing the necessary costs of self-control. To rationalize this ex-ante mechanism, the paper develops an “investor-spender” model in which a relatively patient “investor” allocates resources between liquid and illiquid assets before a more impatient “spender” chooses consumption and credit card borrowing. By placing wealth in less accessible, illiquid assets, the “investor” can preserve long-term savings while limiting the resources available for immediate consumption by the “spender”. This mechanism can generate short-term liquidity shortages and revolving credit use even when households hold substantial assets and report relatively high financial comfort. The model can also reproduce the standard hump-shaped consumption profile of *Convenience users* and the comparatively flatter profile observed among *Revolvers*.

Overall, the empirical patterns documented in this paper suggest that prevailing concerns about *Revolvers* may be incomplete. While the scale of credit card debt in Canada justifies policy attention, treating revolving balances as direct evidence of financial distress relies on assumptions that have received limited empirical scrutiny. The evidence presented here shows that *Revolvers* have comparable levels of financial assets and liabilities to *Convenience users*, do not report higher levels of financial distress, and are more strongly differentiated by liquidity conditions than by income.

Finally, the paper documents a substantial gap between survey and administrative revolving rates. About 21% of DWPT cardholders are classified as *Revolvers*, roughly half the rate reported in credit bureau data (Xiao, 2024). Because this estimate aligns with other surveys (Henry, Shimoda, and Zhu, 2022; Henry, Shimoda, and Rusu, 2024), the gap likely reflects differences type of revolving behaviors captured. In particular, survey data primarily capture persistent revolving, whereas administrative data also include temporary revolving. Accordingly, the paper’s findings should be interpreted as applying primarily to persistent *Revolvers*.

1.2 Related literature

By combining descriptive profiling of *Revolvers*’ and *Convenience users*’ with careful regression analysis of credit card debt determinants, this study contributes to ongoing policy discussions in Canada about the relationship between credit card debt and financial vulnerability. Although some studies address related issues these works either focus on different questions (Bilyk and Peterson, 2015; Allen et al., 2022; Allen, Boutros, and Guttman-Kenney, 2024) or rely on credit report data, which lack key demographic information (Xiao, 2024; Bank of Canada, 2023b). As such, prior research offers limited insight into differences between *Revolvers* and *Convenience users*, particularly with respect to their financial circumstances.

My investigation into the determinants of credit card debt is closely related to the works of Zhou, 2022 and Islam and Picault, 2025, as these studies also estimate the effects of demographic characteristics on revolving behaviors in the Canadian context. Zhou, 2022, for instance, uses the Survey of Financial Security (SFS) to study borrowing and repayment behaviors among older Canadian households. Focusing on individuals aged 55 to 74, the study finds that households with high debt-to-asset ratios and limited holdings of liquid assets face greater difficulty meeting their financial obligations, including credit card repayments. Also using the SFS, Islam and Picault, 2025 investigates credit card repayment behavior and identifies education as the most important factor influencing Canadians’ revolving behavior. My paper distinguishes itself from existing works in two important aspects.

First, because the SFS does not report the exact amount of outstanding balances carried by respondents, data on repayment behavior are limited to broad categories (e.g., paying in full versus carrying a balance). Without data on the amount of outstanding balances, previous papers (Islam and Picault, 2025; Zhou, 2022) are constrained to examining factors influencing the extensive margin of credit card debt. This paper, on the other hand, analyzes both the extensive and intensive margins, thereby offering a more complete perspective on consumers’ borrowing behaviors. Although studies with similar scope exist for other markets (Kim and DeVaney, 2001; Del-Rio and Young, 2005; Lin et al., 2019), to the best of my knowledge, this analysis is among the first to explore both margins in the Canadian context. Beyond documenting the average determinants of credit card debt, my paper also uncovers important life-cycle heterogeneity in these relationships. In particular, I show that the effect of education on the likelihood of revolving

is substantially stronger among working-age cardholders and diminishes after retirement.

Second, this paper incorporates a broader set of robustness checks to strengthen the credibility of the findings. For the extensive margin, it tests the robustness of the estimated results under alternative distributional assumptions for the error term, rather than relying solely on the probit model (Islam and Picault, 2025; Zhou, 2022). Since outstanding balance is observed only for Revolvers, prior works in this literature often analyzes the intensive margin using the Heckman method (Heckman, 1979) to correct for selection bias (Kim and DeVaney, 2001; Lin et al., 2019). These studies nevertheless often lack of good instrument that affects only the probability of revolving, but the amount of outstanding balance conditional on revolving (Del-Rio and Young, 2005). Consequently, identification relies heavily on restrictive functional-form and distributional assumptions concerning the relationship between the unobserved determinants of the extensive and intensive margins (Koop and Steel, 2001).

This paper extends the existing literature on the determinants of credit card debt by varying the degree of correlation the two margins. In particular, I re-estimate the intensive-margin relationships between revolving debt and demographics under over a range of assumed correlations between the unobserved determinants of selection (i.e., revolving status) and outcome (i.e., outstanding balances) equations (Altonji, Elder, and Taber, 2005). While this approach does not resolve the lack of exclusion restriction, it nevertheless illustrates the robustness of estimated relationships to different degree of selection on unobservable (Oster, 2019).

A related literature examines the relationship between revolving credit card debt and household liquidity. The literature documents the “credit card debt puzzle”, whereby some households carry high-interest credit card debt while simultaneously holding liquid assets that could be used to repay them (Gross and Souleles, 2002). One explanation is that households preserve liquid balances to meet expenses for which credit cards cannot be used, making savings and available credit imperfect substitutes (Telyukova, 2013; Greene and Stavins, 2023). Alternative explanations emphasize self-control as reasons why consumers may jointly hold revolving debt and savings (Bertaut, Haliassos, and Reiter, 2009; Gathergood and Weber, 2014; Medina and Pagel, 2025).

Relative to this literature, my paper seeks to explain why revolving and non-revolving cardholders exhibit similar balance-sheet characteristics and

levels of financial well-being, despite *Revolvers* facing seemingly tighter liquidity constraints. The “investor-spender” framework introduced in Section 6 builds upon the “accountant-shopper” model, in which the patient “account” choose not to repay existing credit card debt to enforce tighter borrowing limit on the impatient “shopper” (Bertaut, Haliassos, and Reiter, 2009). Applying this logic, my model interprets the greater concentration of wealth to illiquid assets among *Revolvers* as a mechanism for the patient “investor” to preserve long-term savings by restricting the resources immediately available to the impatient “spender”. This framework can also reproduce consumption and debt dynamics that are qualitatively consistent with observed patterns in the data.

2 Data

The main data source for this paper is the Digital Wallet and Payment Trend Survey (DWPT), a cross-sectional study administered by IPSOS.² As the name suggests, the survey focuses on the landscape of Canada’s payment market, tracking emerging trends and providing detailed insights into credit card ownership, usage patterns, revolving status as well as amount of outstanding balances conditional on revolving.

For each respondent, the survey contains a rich set of demographic information such as age, household income, financial assets and liabilities, among others. The DWPT offers detailed views of respondents’ card portfolio and usage. For each participant, the survey reports the spending and amount of outstanding balances. The breadth and depth of information available allows me to analyze differences between groups of credit card users without having to combine multiple data sources.

For example, one of the most extensively studied datasets of this kind, the Consumer Credit Panel (Lee and Klaauw, 2010; Fulford and Schuh, 2023; Fulford and Schuh, 2024; Haughwout et al., 2024), provides account-level data for the universe of credit card accounts in the U.S; but it captures only a snapshot of the total credit balance for each account. Because the total credit balance consists of both newly incurred spending (i.e., transacting balances) and revolving balances from previous periods, determining an individual’s

²IPSOS is a multinational firm specializing in market research across different industries.

revolving status represents a nontrivial task even with panel data (Fulford and Schuh, 2023).

To illustrate this challenge, consider a case in which an analyst observes that an individual has a total credit card balance of \$1,500 in a given period. This balance could reflect purchases made during the current billing cycle that will be fully paid off before interest is charged, in which case the individual would be a *Convenience user*. Alternatively, some portion of the balance could reflect unpaid debt carried over from a previous period, in which case the individual would be a *Revolver*. Without additional information on credit card spending and repayment behavior, it is difficult to distinguish between these two cases, even when balances are observed over multiple periods.

By contrast, revolving status is readily available in the DWPT. Respondents are first asked, “Do you currently carry an outstanding balance on your card(s)?” Conditional on selecting yes, they are then asked to report the exact dollar amount of their outstanding balance. Based on these responses, the survey defines *Revolvers* as cardholders who carry an outstanding balance, and *Convenience users* as those who do not.

The data used in this analysis include the results from the 2021 and 2022 surveys. To focus on differences between different types of credit card users, I will limit the samples to individuals who own and report positive usage on at least one credit card. This leaves me with 12056 respondents equally divided between the two sample periods.

2.1 Sample Representativeness and Summary Statistics

Table 1 summarizes the demographics of the DWPT sample. Overall, I find the sample here to be representative of all credit card users in Canada in term of age and income (Xiao, 2024).^{3,4} Most credit card users own two cards, with roughly 20% of all users owning three. With 82% of population

³Technically, the DWPT reports only respondents’ income brackets, not their exact income values. However, because these brackets are by a CAD 5000 increment (Appendix A.1), I will use the midpoint of each bracket and treat income as a continuous variable

⁴Cardholders are approximately nine years older than the average Canadian population, primarily as one has to be at least 18 years old to be eligible for a credit card.

owning at least one credit card (World Bank, 2021), there are more cards in circulation than there are Canadians.

Given their popularity, it comes as no surprise that credit cards are the most used method of payment by value share at 61.9%, far outpacing debit cards and cash at 21.6% and 9.7%, respectively. These figures are consistent with reports from the Bank of Canada using the Method of Payment Survey, henceforth MPS (Henry, Shimoda, and Rusu, 2024).

Table 1: Summary statistics of selected sample.

DWPT definition: respondents carrying positive amount of outstanding balances

Alternative definition: respondents who pay less than the entire balance are classified as *Revolvers*

Variable	Mean	Std	Median	75th	80th	90th	95th
% of expenses :							
Cash	9.7	15.1	5	10	10	25	45
Debit	21.6	27.8	10	40	50	70	80
Credit	61.9	32.6	75	90	95	99	100
Other methods	6.8	13.6	0	10	10	20	40
No of credit cards :	1.8	.77	2	2	3	3	3
Monthly expense (\$)	2688	8579	1666	3000	3333	5000	7000
Revolving balance (\$)	895.5	3264	0	0	116.7	2000	5500
<i>Share of Revolvers (%)</i>							
DWPT definition	21.67						
Alternative definition	22.26						
Demographics:							
Age	50.6	16.3	52	64	66	72	75
Share with Bachelor degree (%)	46.4						
Share of Married (%)	.62						
Pretax Income (CAD/year)	84,277	49,756	80,000	100,000	125,000	150,000	200,000
Gender (Male =1)	.49						
Observations	12056						

2.2 Persistent vs Transitory *Revolvers*

Measurement Gaps Between Surveys And Administrative Data: Under the DWPT’s definition, roughly 21% of credit card users are *Revolvers*, less than half the 46% reported in studies utilizing administrative credit bureau data (Xiao, 2024; Allen, Boutros, and Guttman-Kenney, 2024).

Discrepancies in measured credit card borrowing across data sources are well documented across markets. In Canada, the MPS reports a comparable revolving rate of 26% (Henry, Shimoda, and Rusu, 2024).⁵ Similar discrepancies also appear in the U.S. market. Account-level data on balances and repayment behavior indicate that 66% of U.S. credit card users are *Revolvers* (Grodzicki and Koulayev, 2019). By contrast, the Survey of Consumer Payment Choice (SCPC), which applies a definition of revolving similar to those used in the DWPT and MPS, reports a substantially lower rate of 45% (Federal Reserve Bank of Atlanta, 2023).⁶ Although these estimates may not be directly comparable due to differences in unit of observation and survey wordings, they illustrate a broader pattern: survey-based measures tend to produce lower revolving rates than administrative account-level data.

These discrepancies are likely attributable to how revolving status is operationalized. Because credit reports (i.e., TransUnion) allow for the observation of monthly balances and repayments, researchers often define *Revolvers* as anyone failing to fully repay their balance for more than two consecutive months (Grodzicki and Koulayev, 2019; Xiao, 2024). This approach captures both persistent *Revolvers*, who regularly carry balances, as well as transitory *Revolvers*, who may carry balances only temporarily following unexpected shocks. By contrast, survey-based measures such as the DWPT rely on respondents’ self-reported revolving status. In the DWPT, 75% of users identified as *Revolvers* also report that they “usually” repay less than the full amount (Figure 10, Appendix A.2).⁷ This pattern suggests that the

⁵The MPS identifies a revolver as any respondent who indicates that they “had a balance owing last month and did not pay off the full amount.” Notably, this definition does not require respondents to report the exact amount borrowed.

⁶Note that Grodzicki and Koulayev, 2019’s analysis is at account level whereas the SCPC’s unit of observation is individual. Part of the difference in revolving rate may be due one individual borrows on multiple accounts.

⁷The remaining 25% of *Revolvers* may include respondents with temporary borrowing episodes, or measurement error between self-reported outstanding balances and usual repayment habits. Demographically, these cardholders are more similar to *Convenience users* than the “consistent” *Revolvers*.

DWPT is more likely to capture persistent revolving behavior, while less salient borrowing episodes may go unrecorded.

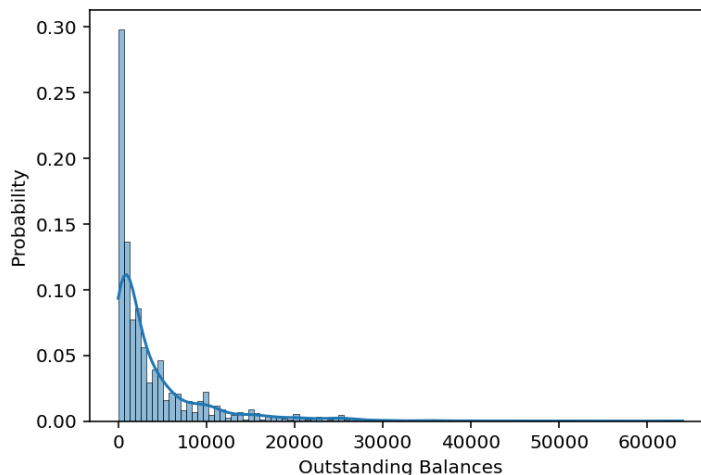
A complementary explanation for the relatively low revolving rate reported in the DWPT is the systematic underreporting of debt in survey data (Brown et al., 2011). Because such underreporting is often unintentional (Zinman, 2009; Akana and Lee, 2025), consumers who revolve only occasionally may fail to recall these transitory debts due to their lower salience. This suggests that larger and more persistent debts are more likely to be salient to respondents and therefore more accurately reported. In accordance with this interpretation, “consistent” *Revolvers*, whose self-reported revolving status and repayment habits align, report significantly higher outstanding balances than those with inconsistent profiles (Figure 1).⁸

Taken together, these measurement differences imply that the results in this paper apply most directly to persistent *Revolvers*, and may not fully generalize to transitory *Revolvers*.

Scale Of Credit Card Debt: Figure 1 and the accompanying table describe the distribution of outstanding balance conditional on revolving. The distribution features a heavy right-tail, highlighting significant debt burdens for a subset of *Revolvers*. Whereas the median *Revolver* carries less than CAD 2000 in outstanding balance, about 25% of these users owe more than CAD 5000 in credit card debt.

⁸The respondents are asked “Do you currently carry an outstanding balance on your card(s)?” and requested to enter the dollar amount of such balances.

Figure 1: Distribution of outstanding balance - *Revolvers* only



Source: Survey question “Do you currently carry an outstanding balance on your card(s)?”

Notes: Consistent *Revolvers* are respondents who carry an outstanding balance and did not select “Entire balance owing” when asked, “Do you usually pay the entire amount owing, pay the minimum amount owing, or somewhere in between?”.

Inconsistent *Revolvers* are respondents who carry an outstanding balance but selected “Entire balance owing” in response to the same question.

Outstanding balance (CAD)	Mean	Stdev	p25	p50	p75	p80	p90
Full sample	868	3180	0	0	0	117.6	2000
<i>Revolvers</i> only	4037	5851	500	1800	5000	6000	10948
Consistent <i>Revolvers</i> only	4751	6169	800	2500	6000	8000	12000
Inconsistent <i>Revolvers</i> only	1854	4122	200	600	1700	2000	4500

3 Portrait of *Revolvers*

This section develops a descriptive portrait of *Revolvers* by comparing their demographic characteristics, balance-sheet positions, and reported financial well-being with those of *Convenience users*. Although *Revolvers* tend to be younger, have lower levels of education, and face tighter liquidity constraints, the two groups report broadly similar financial assets, liabilities, and levels

of financial comfort. These patterns suggest that revolving status alone are not sufficient indicators of financial distress.

3.1 Demographic Profiles And Financial Standings

Demographics: Table 2 highlights differences between *Convenience users* and *Revolvers* across various dimensions. Generally speaking, *Revolvers* are disproportionately younger, have lower income, and are less likely to hold a bachelor’s degree. On average, *Revolvers* make about CAD 10,000 less in pretax income compared to *Convenience users*. As will be evident later on (Table 4), observed differences in income likely reflect correlation between this variable and other demographics, rather than a direct relationship between income and revolving status.

There is also substantial difference in education level between *Revolvers* and *Convenience users*: 35.4% of the former have a bachelor’s degree versus 49.5% of the latter. Since education is strongly correlated with financial sophistication (Lusardi and Mitchell, 2014; Lusardi and Tufano, 2015), this gap might point to lower average financial literacy among *Revolvers*. While this is a possibility, the available evidence suggests that any such difference in financial management skill does not translate into economically meaningful disparities in either balance-sheet positions.

Balance-sheet measures: Figure 2 documents the cumulative and density distributions of financial assets and liabilities among *Revolvers* and *Convenience users*, respectively. Since participants select values from a set of predefined ranges that best reflect their situations, the data on these variables are categorical. I employ the Mann–Whitney U test to verify whether *Revolvers*’ responses differ in a statistically significant manner from those of *Convenience users*. This test was chosen in place of the more commonly used Kolmogorov–Smirnov (KS) due to the categorical nature of the data. One of the key assumptions of the KS test is that the underlying distribution is continuous and not discrete. In situations where the distribution’s support is sparse the KS test is not suitable.⁹

Figures 2a and 2b show that *Revolvers* report slightly higher levels of both financial assets and liabilities than *Convenience users*. Although these distributional differences are statistically significant for assets ($p = .0003$)

⁹While KS test can be modified to for discrete distributions popular implementations of the test often assume continuity

and liabilities ($p = .002$), their magnitudes are economically negligible.

Because balance sheet measures are reported in discrete bins, the data may admittedly obscure finer differences in financial assets and liabilities level between revolving and non-revolving cardholders. Nevertheless, the broadly similar distributions suggest that any such differences are unlikely to be substantial and challenge the notion that *Revolvers* lack the financial buffers needed to withstand unexpected economic shocks.

Around 35% of *Revolvers* report having “No debt” when asked: “Which of the following categories best describes your household’s TOTAL FINANCIAL LIABILITIES?, including ... outstanding balance on credit cards” (Figure 2b).¹⁰ Since these consumers, by virtue of being *Revolvers*, must carry outstanding balances on their credit cards, this most likely reflect measurement error. To verify that the misreporting is not systematic, I conduct a balance test using demographic variables, including income, age, and education level as covariates (Table 7a, Appendix A.2). The results suggest misreporting of debt level is not strongly associated with the observed demographic variables, although systematic differences along unobserved dimensions cannot be ruled out. While such measurement errors would inevitably underestimate *Revolvers*’ levels of liabilities, its overall impact is limited. Given that 90% of *Revolvers* carry less than CAD 10,000 in credit card debt (Figure 1), the resulting distortion is unlikely to meaningfully alter our conclusions.

As a final note, it is unlikely that misreporting of this kind is unique to *Revolvers*. In fact, given that about 70% of all Canadians having some forms of debt (Statistics Canada, 2017) it appears unlikely that more than one-third of *Convenience user* truly have no financial liabilities. What matters for the interpretation of the result is whether this misreporting is concentrated disproportionately among one group. The available evidence does not suggest that this is the case. Conditional on selecting “No Debt”, the demographic differences between *Convenience users* and *Revolvers* (Table 7b, Appendix A.2) closely resemble those reported for the full sample in Table 2.

Assuming consumers prioritize mortgage payments over consumption, mortgages represent financial obligations that restrict the amount of disposable income available for other uses. In that sense, the higher likelihood of being a mortgagor among *Revolvers* likely reflects more restricted liquid-

¹⁰This result persist under the alternative definition of *Revolvers* as respondents who usually do not pay their “Entire owing balance”, albeit at a smaller scale (26%)

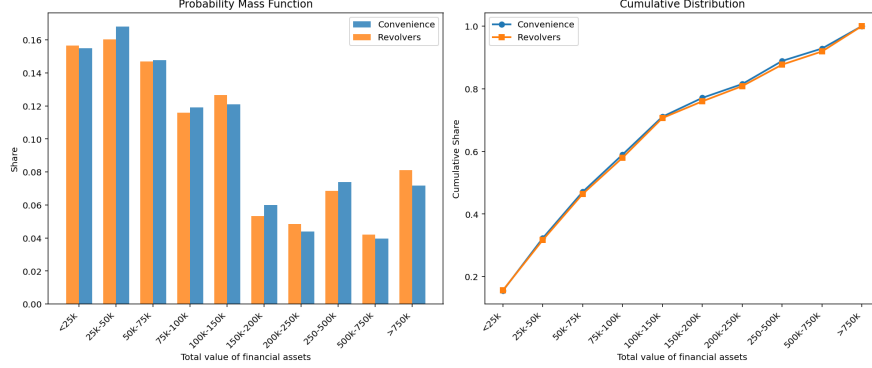
ity constraints within this subset of cardholders. Similar logic applies to their lower rate of investment account ownership. Since these accounts are typically associated with more liquid assets (i.e., stocks, bonds, etc.) lower ownership among *Revolvers* suggests that these cardholders might allocate more of their wealth toward illiquid assets and therefore facing tighter liquidity constraints.

Table 2: Demographics of *Convenience users* and *Revolvers*

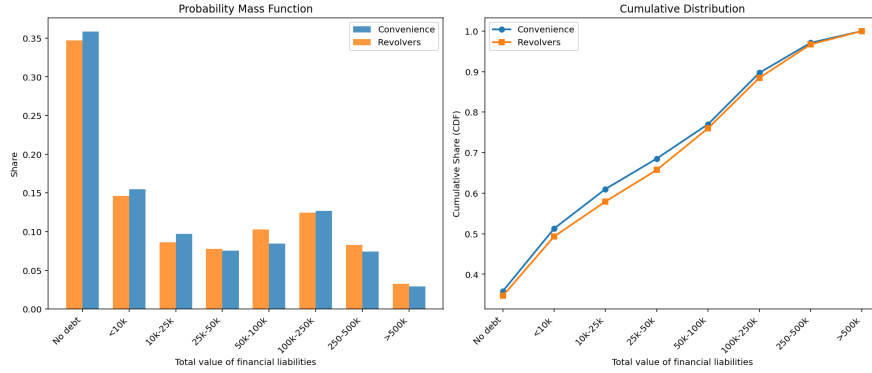
Married Individuals includes both married couples and common law partnership.
 After-tax income is approximated from reported pretax income using federal and provincial tax schedules. The calculation does not account for other tax credits.

Variable	<i>Convenience users</i>	<i>Revolvers</i>
Pretax Income (CAD/year) (\$)	86,696	75,545
After-tax Income (CAD/year) (\$)	61,573	54,377
Age	51	46
Share with Bachelor Degree (%)	49.5	35.5
Share with Mortgage (%)	31.1	39.9
Share with Investment Acc (%)	63.9	45.5
Share of Male (%)	50	46
Share of Married Individuals (%)	62.5	60.6
Household size (persons)	2.4	2.37
Number of credit cards	1.8	1.82

Figure 2: Cumulative distribution of financial assets and liabilities



(a) The DWPT asks: “*What is the total value of your household’s financial assets?*”



(b) The DWPT asks: “*Which of the following categories best describes your household’s TOTAL FINANCIAL LIABILITIES?*”

3.2 Financial Well-Being

Revolving versus Delinquency: The narrative surrounding credit card debt and financial vulnerability often centers on the relationship between credit card debt and delinquencies on other credit products (Keys, Mahoney, and Yang, 2023; Zhao, Witts, and Xiao, 2026). Evidence from administrative data indicates that among delinquent mortgagors, incidents of delinquency on non-mortgage debt, including credit cards, rise sharply twelve months before the initial missed mortgage payment (Zhao, Witts, and Xiao, 2026). Such a concern, nevertheless, conflates revolving status with delinquency.

This distinction matters because revolving status and delinquency are

conceptually different. A credit card account is delinquent when the cardholder fails to make at least the required minimum payment (Capital One, 2025). A *Revolver* may continue making payments above the minimum while carrying part of the balance forward. Thus, although delinquent cardholders are *Revolvers*, most *Revolvers* are not delinquent. In fact, administrative data indicate that the 30-plus-day credit card delinquency rate within the sample period was exceptionally low at approximately 1.47% (DBRS Morningstar, 2023), accounting for merely 3% of all *Revolvers*.¹¹ Furthermore, correlation between mortgage and non-mortgage delinquencies is not unique to credit cards and extends to other products such as home equity line of credit (Chan et al., 2016). This broader pattern indicates that financial distress manifests as delinquencies across multiple credit products, rather than revolving credit use alone.

Subjective Measure Of Financial Comfort: To assess whether revolving debt is directly linked to financial distress, I compare the self-reported financial comfort levels of *Revolvers* and *Convenience users*. If revolving balances are a reliable indicator of financial vulnerability, then conditional on income and wealth, *Revolvers* should report lower perceived financial comfort than their non-revolving counterparts.

Figure 3 plots the distribution of participants’ responses to the question: “How comfortable are you managing your current debt load?”. Since this question was optional and asked only to respondents who did not select “No debt” when reporting their financial liabilities, the sample size is limited to fewer than half of all respondents (5722 observations). Of these missing observations, 66.7% are due to participants answering “No debt”, while the remaining 33.3% are due to participants skipping the question.

Conditional on selecting “No debt” we do not observe any significant differences between this subsample and the overall DWPT sample (Table 7b, Appendix A.2). Therefore, selection into this subgroup, or equivalently, non-observation of comfort levels due to selecting “No debt”, is most likely not systematically related to the observed characteristics. Among those who did not pick “No debt” when asked about their financial liabilities, balance tests show similar characteristics across most variables, although respondents who skipped the question are slightly younger (Table 8, Appendix A.2).¹² Taken

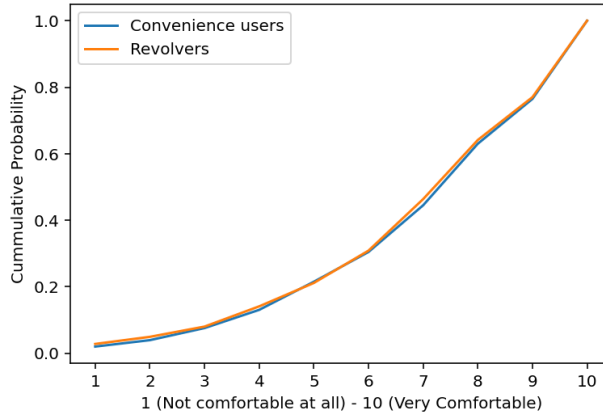
¹¹This proportion is calculated using the 46% overall revolving rate commonly observed in credit bureau data.

¹²Recall that 35% of *Revolvers* misreport their total financial liabilities as “No debt”.

together, these findings suggest that observable selection into the financial-comfort sample is limited and unlikely to materially affect the results.

In line with earlier analysis of financial assets and liabilities, differences in responses between *Revolvers* and *Convenience users* are tested using the Mann–Whitney U test. Overall, I find no significant difference in reported financial comfort levels between *Revolvers* and *Convenience users* (p-value = 0.12). This suggests that *Revolvers* are not experiencing financial distress at a higher rate than *Convenience users*.

Figure 3: The DWPT asks: “How comfortable are you managing your current debt load?”



Regression Analysis: Because comfort is subjective, *Revolvers* may report similar levels of comfort as *Convenience users* even if objective indicators, such as income, assets, or liabilities, suggest weaker financial positions.

To account for this possibility, I estimate the following ordinal regression model of financial comfort:

$$U_i = \delta_0 + \delta_1 \mathbf{1}_{Revolve} + \mathbf{Z}_i \boldsymbol{\beta} + \epsilon_i, \quad (1)$$

where U_i is a latent variable measuring individual’s comfort level, $\mathbf{1}_{Revolve}$ is an indicator function for revolving status and $\mathbf{Z}_i$ is a battery of control variables including age, income, financial assets and liabilities.

Cardholders report a comfort level according to the following rule:

$$Comfort_i = \begin{cases} 1 & \Leftrightarrow U_i \leq \mu_1, \\ 2 & \Leftrightarrow \mu_1 < U_i \leq \mu_2, \\ \vdots & \\ 10 & \Leftrightarrow \mu_9 < U_i \end{cases}$$

where $Comfort_i$ is individual i 's response to the questions “How comfortable are you managing your current debt load?”.

Let $U(\mathbf{Z}_i)$ denotes the latent index conditional on balance-sheet measures and other demographics. *Revolvers*' comfort level is determined by:

$$Comfort_i^{Rev} = \begin{cases} 1 & \Leftrightarrow U(\mathbf{Z}_i) + \delta_1 \leq \mu_1 \Leftrightarrow U(\mathbf{Z}_i) \leq \tilde{\mu}_1 \\ 2 & \Leftrightarrow \tilde{\mu}_1 < U(\mathbf{Z}_i) \leq \tilde{\mu}_2, \\ \vdots & \\ 10 & \Leftrightarrow \tilde{\mu}_9 < U(\mathbf{Z}_i) \end{cases}$$

where $\tilde{\mu}_k = \mu_k - \delta_1$.

Thus, δ_1 captures differences in how *Revolvers* perceive financial comfort relative to *Convenience users*, conditional on the same objective financial indicators. Specifically, a statistically significant $\delta_1 < 0$ raises the effective thresholds faced by *Revolvers*, making them more likely to report a lower level of financial comfort for the same vector of observables $\mathbf{Z}$. Conversely, $\delta_1 > 0$ lowers these thresholds, suggesting that *Revolvers* map the same objective financial state into higher categories of financial comfort.

Results: Table 3 summarizes the results of this exercise. δ_1 is statistically insignificant across all specifications, suggesting no systematic difference in comfort levels between *Revolvers* and *Convenience users*. As a robustness check, I also use double machine learning to allow for more flexible, potentially nonlinear relationships between financial comfort and the observed balance-sheet and demographic characteristics (Chernozhukov et al., 2018; Bach et al., 2022). The resulting estimate remains statistically insignificant (Appendix A.3, Table 11).

The similarity in reported financial standing and comfort between *Revolvers* and *Convenience users*, despite tighter liquidity constraints among the former, may reflect two mechanisms. Credit cards may serve as ex-post liquidity buffers that help constrained consumers absorb temporary cash-flow

shortfalls, or revolving may arise as a cost of holding wealth in less accessible assets for self-control. Section 6 expounds on these possibilities.

Table 3: Ordinal Regression for *Comfort*

Note: The Linear Probability Model (LPM) treats *Comfort* as continuous and uses robust standard errors.

$I(Mortgage)$, $I(Invest)$, $I(University)$ indicate ownership of a mortgage, an investment and a bachelor's degree, respectively.

Variable	Probit	Logit	LPM
$I(Mortgage)$	.020 (.030)	.034 (.051)	.063 (.067)
$I(Invest)$	.238*** (.029)	.397*** (.050)	.558*** (.066)
$I(University)$	-.038 (.029)	-.060 (.049)	-.093 (.064)
Age	.001 (.001)	.001 (.001)	.001 (.002)
Annual Income (\$10k)	-.006* (.003)	-.009* (.005)	-.013** (.007)
<i>Revolver</i>	.001 (.034)	.013 (.058)	.004 (.077)
Assets	Y	Y	Y
Liabilities	Y	Y	Y
Observations	5722	5722	5722
Log-Likelihood	-11744	-11744	
R ²			.017

4 Determinants Of Revolving Behaviors

In this section, I estimate the impacts of income and education attainment, among other factors, on consumer borrowing behaviors. Following prior studies (Kim and DeVaney, 2001; Del-Rio and Young, 2005; Lin et al., 2019), I analyze both the extensive (i.e., whether to revolve) and intensive (i.e., how much to borrow) margins of credit card debt.

4.1 Determinants of Revolving Status

To understand factors influencing consumers' decisions to revolve on their credit cards, I estimate a latent utility model. Let z_i denote the net benefits

of revolving to individual i . z_i is given as:

$$z_i = \alpha_0 + X\alpha_1 + D\alpha_2 + Z_2\alpha_3 + \epsilon_i, \quad (2)$$

where $X = \{Age, Age^2, interest, Income, Income^2\}$. The higher-order terms $\{Age^2, Income^2\}$ are included to capture potential non-monotonic relationships between *Age*, *Income* and propensity to revolve, as suggested by results of the clustering analysis (Table 12, Appendix A.5). The vector $D = \{Invest, Mortgage, University\}$ contains indicator variables for whether individual i has an investment account, a mortgage, or a bachelor's degree, respectively. As identified in the clustering analysis (Appendix A.5), the effect of education on revolving status appears to bifurcate around retirement age. To allow for such possibility, I also estimate an alternative specification of Eq (2) that includes the interaction term $University * \mathbf{1}_{Over\ 65}$ and $\mathbf{1}_{Over\ 65}$ as additional regressors. Finally, Z_2 represents a set of categorical control variables, including reported levels of financial assets and liabilities, province of residence, as well as PRIZM segmentation codes.^{13 14}

The model can be estimated using either probit or logit regression, depending on the distributional assumptions regarding the error term ϵ_i . To assess the robustness of the results to these assumptions, I also estimate a linear probability model (LPM) version of Eq (2), using revolving status as the dependent variable. The results from the logit and LPM specifications are broadly consistent with those of the main probit model and are presented in Appendix A.4 (Table ??).

4.2 Determinants of Outstanding Balance

As *Balance* is observed only for *Revolvers*, estimating the intensive-margin equation by OLS on this selected subsample may produce selection-biased estimates (Kim and DeVaney, 2001). A common approach to account for this bias is to employ the two-stage Heckman method (Heckman, 1979). This method involves estimating a probit specification of Eq (2) in the first

¹³PRIZM is customer segmentation tool used in marketing researches. It classifies customers into 1 of 68 groups reflecting differences in income, life stages, etc...<https://enviroanalytics.com/en-ca/data/segmentation/prizm>. The DWPT assigns respondents into different segments using their survey answers.

¹⁴While a random forest classifier could be used in place of Eq. (2), such models are less interpretable and do not provide the same transparent relationship between observable characteristics and revolving behavior.

stage. In the second stage, researchers can include inverse Mills ratio derived from the first-stage estimates to correct for potential sample-selection bias.

Formally, the estimating equation in the second stage is:

$$\log \text{Balance} = \beta_0 + \log X\beta_1 + D\beta_2 + Z_2\beta_3 + \rho\sigma_\nu \frac{\phi(X, D, Z_2)}{\Phi(X, D, Z_2)} \quad (3)$$

where X, D, Z_2 are as described in Eq (2) and $\frac{\phi(X, D, Z)}{\Phi(X, D, Z)}$ is the Inverse Mills ratio (IMR) calculated using the probit specification of Eq (2).

Barring scenarios where issuers purposefully limit access to credit for specific subgroups of consumers,¹⁵ it is difficult to identify demand-side factors that affect only the decision to revolve but not the amount of outstanding balance carried (Del-Rio and Young, 2005). Consequently, I do not have a credible exclusion restriction for the selection equation. Identification of the Heckman specification therefore relies primarily on the nonlinearity of the inverse Mills ratio and on the assumed joint normality of the selection- and outcome-equation errors. Estimates may be sensitive to violations of these functional-form and distributional assumptions (Koop and Steel, 2001).

As a robustness check, I therefore re-estimate the second-stage equation over a range of imposed values of ρ (Altonji, Elder, and Taber, 2005). For each assumed correlation, I recalculate the selection-adjusted outcome and estimate the corresponding intensive-margin coefficients. This exercise does not identify the true degree of selection on unobservables, but shows how strong the correlation between the unobserved determinants of revolving status and outstanding balances would need to be to materially alter the estimated relationships. Details regarding the implementation of the robustness check can be found in Appendix A.4

4.3 Results

Table 4 summarizes the results of this exercise. For the extensive margin, it reports the average marginal effects of each variable on the probability of revolving. For the intensive margin, following the convention in the macroeconomic literature, I estimate the relationship between debt and other characteristics using a log-log specification (Gourinchas and Parker, 2002; Fulford and Schuh, 2023). Accordingly, β_1 are interpreted as elasticities: each

¹⁵In that case, the dummy for said subgroup will affect only the decision to revolve, through supply-side restrictions, but not amount borrowed, conditional on revolving.

coefficient represents the percentage difference in outstanding balances associated with a 1% difference in the corresponding variable, conditional on the included covariates. For the vector of binary variables, β_2 represents the percentage differences in the amount of outstanding balance among *Revolvers* with and without an investment account, a mortgage and a university’s degree, respectively.

As an additional robustness check, I estimate Eq.(3) using different sets of covariates to assess whether the results are sensitive to this choice.

Given the image of struggling individuals turning to credit card debt to make ends meet, it is striking that income plays only a minimal role in consumers’ borrowing behavior. Controlling for assets and liabilities, a CAD 10,000 difference in pretax income is associated with approximately a 1 percentage point (pp) lower probability of revolving and no statistically significant difference in outstanding balances.

On the intensive margin, I find no robust association between *Income* and outstanding balances. Although the estimated coefficients are marginally significant at the 10% level in some specifications, this significance does not persist across alternative values of ρ (Figure 14).

The observed relationship, or lack thereof, between income and revolving behaviors suggests that the gap in income between revolving and non-revolving cardholders reflects a correlation between this variable and other demographic characteristics, rather than a direct relationship with revolving behavior. Education level is one likely source of this correlation.

Holding a bachelor’s degree (i.e., *University* = 1) is associated with an 8 to 9 percentage point lower probability of revolving (specifications (1) and (2), Table 4). Conditional on revolving, it is also associated with approximately 34% lower outstanding credit card balances (specification (1), Table 5). Although the magnitude of this association varies substantially with the assumed value of ρ , it remains statistically significant across the range of values considered (Figure 14).

Given the positive correlation between education and financial literacy (Lusardi and Mitchell, 2014; Goyal and Kumar, 2021), the negative association between *University* and revolving behavior aligns with the perception of *Revolvers* as financially naive (Han, 2023; Agarwal et al., 2026). Nevertheless, any such differences do not appear to translate into economically meaningful differences in observed balance-sheet outcomes (Figure 2) or self-reported financial comfort (Figure 3) between *Revolvers* and *Convenience users*.

The association between *University* and revolving status also differs above and below age 65 (specification (2), Table 4). Among respondents over the age of 65, holding a bachelor’s degree is associated with a 5pp lower the probability of revolving in the probit model, relative to 9 pp for those under 65. One possible explanation is that older individuals are, on average, more financially sophisticated (Henry, Shimoda, and Zhu, 2022), making educational attainment a weaker proxy for financial literacy at older ages. This pattern also appears in the linear probability and logit models, although the difference across age groups is statistically significant only in the former (Table 4).

Because the DWPT does not directly measure financial literacy, discussions regarding the association between financial sophistication and revolving behaviors remains necessarily tentative.

Conditional on assets, liabilities, and the remaining covariates, investment-account ownership is associated with a 10 pp lower probability of revolving. Interestingly, mortgage ownership exhibits the opposite relationship, corresponding to a 7 pp higher probability of revolving. Controlling for other factors, *Invest* and *Mortgage* serve as proxies for respondents’ liquidity. Specifically, since most financial assets are liquid (i.e., mutual funds and stocks), owning an investment account plausibly reflects a greater allocation of wealth toward more these assets, and hence indicates laxer liquidity constraints. Conversely, mortgages represent financial obligations that must be paid from disposable income, and therefore likely indicate tighter liquidity constraints. The negligible effect of income, combined with the opposing impacts of *Invest* and *Mortgage*, are consistent with liquidity-related factors playing an important role in revolving behavior.

In terms of the intensive margin, *Mortgage* (Table 5) is strongly associated with increases in the amount borrowed. These observations stand in contrast with empirical findings for the U.S (Kim and DeVaney, 2001) and U.K market (Del-Rio and Young, 2005), where ownership of mortgages have no significant impacts on the amount of outstanding balance. Coefficients associated with *Mortgage* remain statistically significant across a wide range of ρ , suggesting that the relationship is robust to selection on unobservables (Figure 14).

Although the coefficients associated with ownership of investment account is statistically insignificant, point estimates remain negative across all specifications (Table 5). This aligns with the hypothesis that *Invest* = 1 indicating stronger liquidity conditions.

As discussed in Section 6, liquidity may shape revolving behavior in two

ways: as an ex-ante self-control mechanism that helps preserve long-term savings, or as an ex-post stopgap for easing liquidity pressure. However, because the DWPT lacks detailed information on respondents' asset composition and direct measures of liquidity, this discussion remains speculative.

Table 4: Extensive Margin: Alternative Specifications

Assets, liabilities, PRIZM, and province controls are included in all specifications.
Interest, *Age*, *Income* enter all specifications in levels.
Some observations are excluded beof missing PRIZM codes.

	Probit ME		Logit ME	LPM
	(1)	(2)	(3)	(4)
<i>Interest</i>	-0.01*** (0.002)	-0.01*** (0.002)	-0.02*** (0.00)	-0.01*** (0.00)
<i>Age</i>	0.01*** (0.002)	0.01*** (0.002)	0.01*** (0.00)	0.01*** (0.00)
<i>Age</i> ²	-0.0001*** (1.55e-05)	-0.0002*** (2.15e-05)	-0.00*** (0.00)	-0.00*** (0.00)
<i>Income</i>	-0.01*** (0.002)	-0.01*** (0.002)	-0.01*** (0.00)	-0.01*** (0.00)
<i>Income</i> ²	0.0002 (0.000)	0.0002 (0.000)	0.00 (0.00)	0.00 (0.00)
<i>University</i>	-0.08*** (0.008)	-0.09*** (0.008)	-0.09*** (0.01)	-0.09*** (0.01)
<i>University</i> $\times$ <i>Age</i> $\geq$ 65		0.04* (0.022)	0.03 (0.02)	0.04** (0.02)
<i>Mortgage</i>	0.07*** (0.008)	0.07*** (0.008)	0.07*** (0.01)	0.07*** (0.01)
<i>Invest</i>	-0.10*** (0.008)	-0.10*** (0.008)	-0.09*** (0.01)	-0.10*** (0.01)
<i>Age</i> $\geq$ 65		0.02 (0.020)		
N	11,554	11,554	11,554	11,554
(Pseudo) R ²	0.08	0.08	0.08	0.08

*p $\leq$ 0.10, **p $\leq$ 0.05, ***p $\leq$ 0.01.

Table 5: Intensive Margin: Heckman Two-Step Correction

The first-stage probit specification and estimated inverse Mills ratio are held fixed across specifications.
 Assets, liabilities, PRIZM, and province controls are included in all second-stage specifications.
 Heteroskedasticity-robust standard errors are reported in parentheses.

	Log Balance				
	(1)	(2)	(3)	(4)	(5)
<i>Mortgage</i>	0.311*** (0.117)	0.303*** (0.105)	0.295*** (0.106)	0.419*** (0.096)	0.363*** (0.094)
<i>Invest</i>	-0.022 (0.136)	-0.043 (0.121)	-0.045 (0.122)	-0.128 (0.106)	-0.151 (0.103)
$\log(\text{Interest})$	-1.274*** (0.365)	-1.331*** (0.342)	-1.297*** (0.344)	-1.591*** (0.321)	-1.549*** (0.314)
$\log(\text{Limit})$	0.184*** (0.033)				0.146*** (0.033)
$\log(\text{Age})$	9.395*** (3.480)	10.082*** (3.116)	9.444*** (3.135)	8.805*** (2.966)	8.746*** (2.822)
$\log(\text{Age})^2$	-1.169** (0.485)	-1.263*** (0.433)	-1.179*** (0.436)	-1.082*** (0.410)	-1.082*** (0.391)
$\log(\text{Income})$	1.424 (1.081)	1.865* (1.039)	1.615 (1.034)	2.095* (1.070)	1.814* (1.037)
$\log(\text{Income})^2$	-0.063 (0.051)	-0.083* (0.049)	-0.071 (0.048)	-0.093* (0.050)	-0.086* (0.048)
$\log(\text{Expenditure})$		0.275*** (0.033)	0.976*** (0.291)		0.918*** (0.285)
$\log(\text{Expenditure})^2$			-0.047** (0.019)		-0.044** (0.019)
<i>University</i>	-0.344*** (0.128)	-0.367*** (0.115)	-0.356*** (0.116)	-0.431*** (0.102)	-0.428*** (0.099)
$\text{Age} \geq 65$				-0.375** (0.154)	-0.297** (0.151)
$\text{University} \times \text{Age} \geq 65$				0.036 (0.242)	-0.092 (0.239)
IMR	0.567 (0.458)	0.623 (0.393)	0.585 (0.400)	1.033*** (0.322)	0.944*** (0.310)
N	2502	2502	2502	2502	2502
F-statistic	3.74	4.37	4.4	3.34	4.66
R ²	0.12	0.138	0.14	0.112	0.148
Adjusted R ²	0.082	0.101	0.103	0.073	0.11

*p ≤ 0.10, **p ≤ 0.05, ***p ≤ 0.01.

5 Consumption Path

The previous sections demonstrated that the prevailing belief of credit card debt as a sign of financial vulnerability is somewhat overstated. While *Revolvers* tend to have lower income and educational attainment compared to *Convenience users*, both groups report comparable balance-sheet positions and financial well-being. Nevertheless, there is indirect evidence supporting the importance of financial literacy in determining consumer borrowing behaviors. In this section, I investigate how *Revolvers* differ from *Convenience users* with respect to their consumption over the life cycle. This analysis builds upon theoretical results of Fulford and Schuh, 2024 regarding the consumption and debt profiles of different groups of credit card users.

Due to data limitations, studies in this literature typically do not observe individuals' revolving status or their credit card debt stocks directly (Gourinchas and Parker, 2002; Fulford and Schuh, 2024). By leveraging the DWPT's detailed information on revolving status, outstanding balances, and payment behavior, this section evaluates the empirical validity of standard model predictions.¹⁶ In doing so, it provides evidence on the mechanisms underlying credit card borrowing and the role of revolving debt in shaping consumption over the life cycle.

5.1 Revolving Debt and Life-Cycle Consumption: Theoretical Predictions

This subsection summarizes the benchmark life-cycle patterns that motivate the empirical analysis. Consider a two-population framework in which consumers differ in patience (Meier and Sprenger, 2010; Fulford and Schuh, 2024). Relatively impatient consumers place greater weight on current consumption, save little, and carry revolving balances on their credit cards. These consumers correspond to *Revolvers*. More patient consumers, by contrast, save part of their income and repay their credit card balances in full each period, corresponding to *Convenience users*.

Under a standard hump-shaped earnings profile (Gourinchas and Parker, 2002; Telyukova, 2013; Fulford and Schuh, 2024), the framework implies different life-cycle consumption patterns across the two groups. Because im-

¹⁶Consumption is not directly observed in the DWPT but can be inferred from credit card spending and share-of-wallet measures.

patient consumers accumulate fewer assets, their consumption should track current income more closely. Accordingly, the consumption profile of *Revolvers* should exhibit a more pronounced hump-shaped pattern than that of *Convenience users*, who can smooth consumption by accumulating assets during their working years and drawing them down later in life

Standard frameworks also provide a benchmark for the age profile of revolving balances. Conditional on revolving, consumers may accumulate balances during the early stages of the life cycle, when expected future income is rising, and reduce them as they approach retirement. A terminal repayment or bequest motive therefore generates a hump-shaped age profile of outstanding balances: debt rises during early working ages, peaks in midlife, and subsequently declines.

For ease of reference, the two benchmark implications are summarized as follows:

Benchmark 1. *Revolvers* exhibit a more pronounced hump-shaped consumption profile than *Convenience users*.

Benchmark 2. Conditional on revolving, outstanding balances rise during the early stages of the life cycle and decline as consumers approach retirement.

These benchmarks have important policy implications. If *Revolvers*' consumption tracks income more closely over the life cycle, they may experience sharper consumption declines during retirement, when earnings fall and repayment obligations remain. The corresponding policy concern is that *Revolvers* may enter retirement with both limited savings and outstanding credit card debt. Policies targeting this group would therefore seek to strengthen financial buffers before retirement, either by encouraging precautionary saving or by reducing the accumulation of high-interest debt earlier in the life cycle.¹⁷

5.2 Constructing A Synthetic Cohort

Because long household panels with detailed consumption and credit card borrowing information are rarely available, the literature often relies on syn-

¹⁷One possible approach is to impose higher minimum-payment thresholds, particularly during prime working ages when income is relatively high.

thetic cohort constructed from cross-sectional surveys (Deaton, 1985; Gourinchas and Parker, 2002; Dang and Lanjouw, 2023). Following this approach, I use repeated cross-sections from the DWPT to construct age profiles of consumption and credit card debt by revolving status.

Let $Y_{i,a}$ denote an outcome of interest (e.g., consumption, outstanding balance) for individual i at age a . The main identifying restriction of the synthetic cohort approach is:

$$E[Y_{i,a+h} \mid Revolve_{i,a} = 1, \mathbf{X}] = E[Y_j \mid A_j = a + h, Revolve_j = 1, \mathbf{X}]$$

where $\mathbf{X}$ denotes the vector of observable characteristics. In other words, the cross-sectional outcomes of older individuals serve as a valid proxy for the future average outcomes of today’s younger consumers, conditional on observables and revolving status. The following assumptions ensure that the economic conditions under which this conditional-mean restriction is plausible.

Assumption 1. *Individuals behave as if there are no major structural changes in the economy. In particular, consumers expect their wages to follow the observed historical patterns, conditional on their characteristics (e.g., age, current income, education, etc...)*

Assumption 2. *Revolving behavior is sufficiently persistent so that older Revolvers are representative of the future average outcomes of younger Revolvers, conditional on observed characteristics.*

Standard life-cycle models of consumption and savings often feature similar versions of Assumption 1, in which agents’ expectations of future income, conditional on current state variables, are time-invariant (Gourinchas and Parker, 2002; Telyukova, 2013; Fulford and Schuh, 2024). Since individuals’ consumption and saving decisions depend on their expectations of future earning, *Assumption 1* and its variations imply that consumers with similar income trajectories and characteristics (e.g., ability, education level, etc.) would exhibit similar consumption dynamics over the life cycle.¹⁸

Empirically, expected income growth among Canadians has remained stable over the last decade (Figure 4), with the notable yet unsurprising exception of the Covid-19 pandemic in 2020, which falls outside the sampling

¹⁸Formally, Assumption 1 requires that $E_t[Income_{t+1}|\mathbf{S}] = E_{t+k}[Income_{t+k+1}|\mathbf{S}]$ where $\mathbf{S}$ denote the vector of state variables. A canonical example satisfying this condition is an AR(1) income process with time-invariant parameters.

periods of 2021 and 2022. Even though expectations among Canadians aged 18 to 24 are more volatile than those of other age groups; young consumers are underrepresented in the DWPT (Table 1) and hence are unlikely to materially affect the results. Overall, Figure 4 provides direct empirical support for Assumption 1.

Whereas Assumption 1 allows us to infer general consumption dynamics, Assumption 2 is necessary to compare the life-cycle consumption profiles of *Convenience users* and *Revolvers* from cross-sectional data. Without imposing stability on group membership (i.e., revolving status), transitions between groups would contaminate the age profiles. By assuming revolving status is sufficiently persistent, we can use the older *Revolvers* in our dataset to proxy the future consumption trajectories of younger *Revolvers*.

While admittedly restrictive, Assumption 2 conforms with both the nature of our data and the theoretical framework underlying Benchmark 1. Importantly, the synthetic-cohort analysis is implemented using one-year increments (i.e., $h = 1$). The assumption therefore requires revolving status to be sufficiently persistent only for older *Revolvers* to approximate the outcomes of younger *Revolvers* one year later, rather than requiring individuals to remain *Revolvers* permanently throughout the life cycle. Empirically, this local persistence condition assumption finds strong support in the fact the quarterly *Revolver-to-Revolver* transition probability remains as high as 0.83 over a 15-year horizon (Fulford and Schuh, 2023).¹⁹ Evidence from the Credit Card Database similarly indicates that a continuous revolving spell lasts approximately ten months on average (Grodzicki and Koulayev, 2019).

As a validation exercise, I apply the same synthetic-cohort procedure to household income, an outcome that is directly observed and whose life-cycle pattern is well established. Under Assumptions 1 and 2, income observations for otherwise comparable individuals of different ages can likewise be stitched together to approximate an age-income trajectory. The resulting profile rises during working ages and declines around retirement, consistent with standard life-cycle income dynamics (Gourinchas and Parker, 2002). The ability of the procedure to reproduce this familiar pattern provides supporting evidence that the estimated consumption and debt profiles capture economically meaningful age dynamics.

It is worth mentioning that while Figure 5 suggests that *Revolvers* earn

¹⁹In other words, a *Revolver* today have a 83% chance of remaining a *Revolver* in the next quarter.

less than *Convenience users* on average, this difference is not statistically significant after conditioning on other characteristics (Table 10).

Figure 4: Expected income growth for the next year by age groups

Source: Data are from Canadian Survey of Consumer Expectations (Bank of Canada, 2023a). The author is responsible for the figure

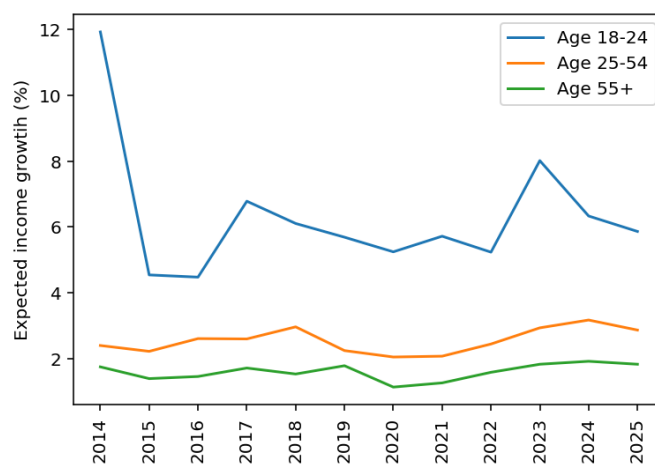


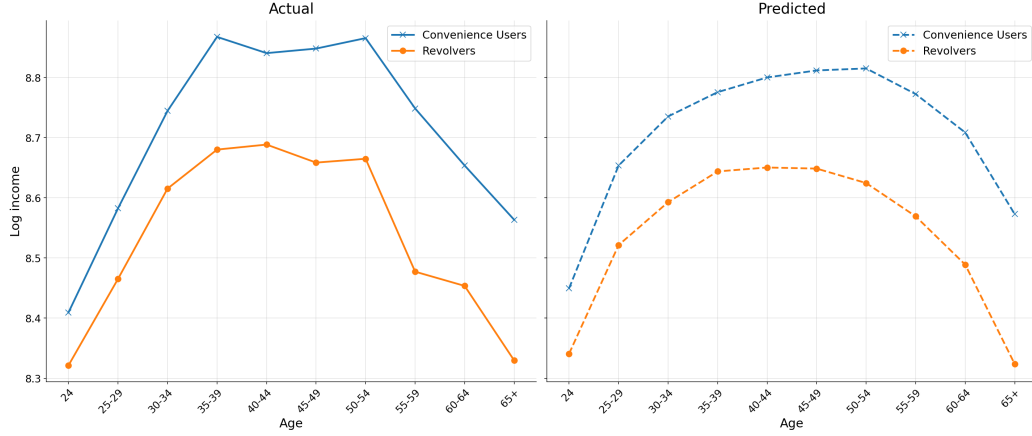
Figure 5: Actual vs. Predicted Income Path

Predicted income is constructed using estimates from the following regression:

$$\log(\text{consumption}_i) = \alpha_1 + \gamma_1 \text{Revolver}_i + \beta \mathbf{Z}_i + \alpha_2 \text{Age}_i + \alpha_3 \text{Age}_i^2 + \gamma_2 \text{Age}_i \cdot \text{Revolver}_i + \gamma_3 \text{Age}_i^2 \cdot \text{Revolver}_i$$

$\mathbf{Z}_i$ includes household size, the number of children, and educational attainment.

Full regression result is reported in Table 10.



5.3 Empirical Patterns

Before presenting the results, it is important to note that the DWPT is a repeated cross-sectional not panel data. Therefore, the estimated profiles are suggestive of life-cycle patterns but should not be interpreted as tracking the same cohorts over time.

Benchmark 1 (Consumption Paths): To test the empirical validity of Benchmark 1, I will estimate the following equation:

$$\begin{aligned} \log(\text{consumption}_i) = & \alpha_1 + \gamma_1 \text{Revolver}_i + \beta \mathbf{Z} \\ & + \alpha_2 \text{Age}_i + \alpha_3 \text{Age}_i^2 \\ & + \gamma_2 \text{Age}_i \cdot \text{Revolver}_i + \gamma_3 \text{Age}_i^2 \cdot \text{Revolver}_i \end{aligned} \quad (4)$$

where *consumption* is average monthly expense and *Age* is the respondent's age. The indicator function *Revolver* equals to 1 if *i* is a *Revolver* and 0 otherwise. Finally, $\mathbf{Z}$ is a vector of control variables consisting of household size, the number of children and education level. Income is intentionally excluded from Eq. (4) so that *Age* captures the underlying income trajectory over the life cycle.

The key coefficients of interest here are: α_2 , α_3 , γ_2 and γ_3 . Whereas α_2 , α_3 measure how consumption levels of *Convenience user* evolve over time, γ_2 and γ_3 capture differences in the consumption trajectory of *Revolvers* relative to that of *Convenience user*.

If Benchmark 1 holds, consumption should increase during the early-to-mid career stage and decline later in life. In the baseline profile (i.e., *Convenience users*'), this requires $\alpha_2 > 0$ and $\alpha_3 < 0$, with the positive early-life effect dominating initially. For *Revolvers*, the hump-shaped pattern should be more pronounced: their consumption should rise more steeply during the early-to-mid career stage and fall more sharply during retirement. Empirically, this implies $\alpha_2 + \gamma_2 > 0$ and $\alpha_3 + \gamma_3 < 0$, with the later-life decline sufficiently large relative to the earlier increase. Table 6 and Figure 6 summarize the main findings of this exercise.

While both *Convenience users*' and *Revolvers*' consumption profiles follow the hump-shaped pattern commonly documented in the literature (Gourinchas and Parker, 2002), the pattern is more compressed among *Revolvers* (Figure 6). *Revolvers* exhibit significantly higher initial consumption levels but experience tempered growth during their prime working years. The two profiles cross at approximately age 50-54. Beyond this point, their trajectories diverge: *Revolvers*' consumption begins a gradual decline, while *Convenience users*' continues to trend upward, leaving the former with lower spending (Figure 6).

While empirical findings reject Benchmark 1, the observed consumption trajectories of *Revolvers* and *Convenience users* remain fundamentally consistent with a two-population model characterized by heterogeneous discount factors, a la Fulford and Schuh, 2024. Suppose both non-revolving cardholders (i.e., patient population) and their revolving counterparts (i.e., impatient population) face identical lifetime resources.

Under the Permanent Income Hypothesis (PIH), any increase in current consumption must be “financed” by a reduction in future consumption. The impatient population prefer to consume more today by revolving on their credit cards, effectively borrow against future earnings. This front-loading creates a higher initial consumption level but leaves a smaller budget for later life. Consequently, their consumption growth slows down earlier and eventually fall below that of *Convenience users*, despite potentially similar income dynamics (Figure 5, Table 10).

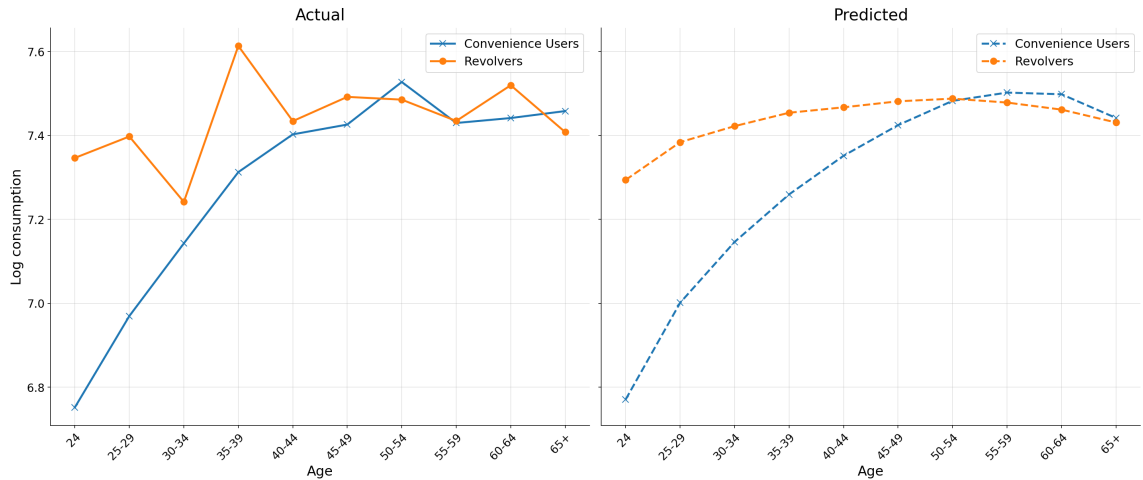
Deviations from Benchmark 1 could arise from the assumption of resource parity. Within the PIH framework (Friedman, 1957), lifetime budget com-

prises both wage income and returns from financial assets. Because the standard model predicts that revolving users do not save due to their impatience (Angeletos et al., 2001; Fulford and Schuh, 2024), this group is expected to possess smaller lifetime budgets relative to the non-revolving/patient population. Empirically, however, both revolving and non-revolving cardholders report similar levels of financial asset holdings and liabilities (Figure 2).

Section 6 puts forth a model in which *Revolvers* behave patiently when making investment decisions but impatiently when choosing current consumption. The model can reconcile deviations from the standard model and reproduce the broad consumption dynamics observed among revolving and non-revolving cardholders.

Benchmark 2 (Debt Dynamics): Benchmark 2 is broadly consistent with results from our investigation of the intensive margin of credit card debt (Table 5). Outstanding balances increase in early stages ($\beta_{age} = 9.395$, specification (1), Table 5) as individuals postpone repayment to prioritize current consumption. Nevertheless, as people want to avoid leaving negative inheritances, *Revolvers* start to repay their owing balances during later stages ($\beta_{age^2} = -1.169$), resulting in a decrease in outstanding balance. This pattern is robust the varying degree of correlation between the selection and outcome equations (Figure 14).

Figure 6: Actual vs. Predicted consumption path



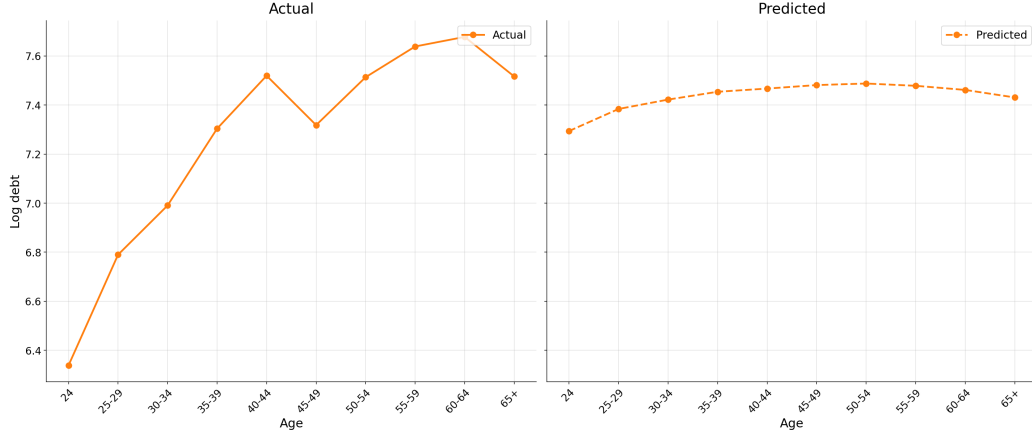
This figure plots the mean predicted value of log *consumption* by age cohorts.

Table 6: Consumption Dynamics

	log(consumption)
Revolver	1.234*** (.211)
Age	.057*** (.004)
Revolver $\times$ Age	-.038*** (.009)
Age^2	-.0005*** (.00003)
Revolver $\times Age^2$	.0003*** (.00009)
University	.168*** (.017)
Children	-.004 (.014)
Household Size	-.007 (.009)
Constant	5.691*** (.092)
R^2	.050
Observations	12,056

Notes: Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 7: Actual vs. Predicted debt dynamics



This figure plots the mean predicted value of $\log \text{Balance}$ by age cohorts.

6 Liquidity, Revolving Debt And Financial Well-Being

As discussed at the beginning of Section 3, *Revolvers* appear to experience more binding liquidity constraints, as reflected by their higher likelihood of being mortgagors. This raises a natural question: how do *Revolvers* enjoy comparable levels of financial well-being relative to Convenience users despite possessing less financial flexibility? This seeming contradiction motivates a closer exploration into the role of credit cards as a source of credit.

This section presents two possible explanations for the relationship between liquidity, revolving debt, and financial well-being: an ex-post perspective centered on access to liquidity (Telyukova, 2013), and an ex-ante perspective treating revolving debt as a necessary costs of self-control.

6.1 Credit Cards As Ex-Post Remedy

In comparison to alternatives such as personal loans and lines of credit, which typically require a relatively lengthy and more demanding application process, credit cards present a much more accessible source of funds. The ubiquity of pre-approved credit card offers means that for some consumers access to credit is just a click away. Much like mortgage payments reduce dispos-

able income, the option to revolve, or equivalently not fully repay credit card balances, expands the pool of liquid funds available to *Revolvers*.

Thus, credit card debt affords consumers the ability to smooth out any shocks while simultaneously keeping up with other financial obligations, such as mortgage payments (Greene and Stavins, 2023). Through this lens, unsecured credit provided by credit cards allows *Revolvers* to sustain the same financial comfort as *Convenience users* despite tighter liquidity constraints. The connection between revolving balances and demand for liquid assets is not a new idea and has been used to explain the “credit card debt puzzle”, where individuals borrow at high interest rate from their credit cards while at the same time holding low-yielding liquid assets.

6.2 Ex-ante Self-Control

Theoretical Model: An alternative explanation for the relationship between revolving debt, liquidity, and financial well-being posits that *Revolvers* exhibit time-inconsistent preferences (i.e., shifting discount factors) across investment and consumption decisions.

Consider the following model in which each period consists of an investment and a consumption subperiod. Decisions are sequential: agents first decide how much to invest in liquid and illiquid assets. Subsequently, agents consume, potentially by borrowing on credit cards, taking both the remaining disposable income and the asset allocation as given. Because *Revolvers* exhibit inconsistent time preferences, these cardholders behave as patient “investors” during the investment phase and as impatient “spenders” during the consumption phase.

For expositional clarity, consider a simplified two-period version of the model. Taking the “spender”’s decisions as given, we focus on the “investor”’s problem:

$$V = \max_{L,K} U(C_1) + \beta_I \mathbb{E}[U(C_2)] \quad (5)$$

subject to

$$\begin{aligned} C_1 &= y_1 - L - K + \theta_L L + \theta_K pK, \\ C_2 &= y_2 + (1 - \theta_L)L + (1 - \theta_K)pK, \end{aligned}$$

where y_t denotes exogenous income, while L_t and K_t represent investments in liquid and illiquid assets, respectively. Finally, $p < 1$ reflects the fact that illiquid assets cannot be readily liquidated.

Because the “spender” may liquidate either type of assets to finance first-period consumption, only the remaining fractions, $1 - \theta_L \geq 0$ and $1 - \theta_K \geq 0$, are carried into period 2. Hence, the “investor” takes θ_L, θ_K as given when making investment decisions.

For both assets to be held in strictly positive amounts, the first-order conditions must hold with equality:

$$\begin{aligned}\frac{\partial V}{\partial L} &= -(1 - \theta_L)U_{C_1} + \beta_I(1 - \theta_L)U_{C_2} = 0 \implies U_{C_1} = \beta_I U_{C_2}, \\ \frac{\partial V}{\partial K} &= -(1 - p\theta_K)U_{C_1} + \beta_I(1 - \theta_K)pU_{C_2} = 0 \implies (1 - p\theta_K)U_{C_1} = \beta_I(1 - \theta_K)pU_{C_2}.\end{aligned}$$

This requires

$$\frac{(1 - \theta_K)p}{1 - \theta_K p} = 1$$

Thus, when $p < 1$, so that liquidating the illiquid asset entails a penalty, consumers will only hold liquid assets.

However, if the “spender” is sufficiently impatient to exhaust all liquid assets and liquidate the maximum accessible share of illiquid assets, then $\theta_L = 1$ and $\theta_K = \bar{\theta}_K < 1$. This changes the investor’s budget constraints to:

$$\begin{aligned}C_1 &= y_1 - K + \bar{\theta}_K p K + L - L, \\ C_2 &= y_2 + (1 - \bar{\theta}_K)pK.\end{aligned}$$

In this scenario, since all liquid assets get liquidated by the “spender”, the “investor” uses the liquidity friction of the illiquid asset as a commitment device to maintain long-term savings. While the resulting constraint on disposable resources may lead the “spender” self to revolve on credit cards, the “investor” self tolerates this borrowing cost as the necessary price for preserving long-term wealth.²⁰

The model therefore explains the observed differences in asset composition between revolving and non-revolving cardholders. By investing in illiquid

²⁰Let $b(y_1, y_2, K)$ denote the borrowing chosen by the “spender” conditional on income and the investor’s allocation to illiquid assets. Under this scenario, the budget constraints become

$$\begin{aligned}C_1 &= y_1 - K + \bar{\theta}_K p K + b(y_1, y_2, K), \\ C_2 &= y_2 + (1 - \bar{\theta}_K)pK - (1 + r)b(y_1, y_2, K).\end{aligned}$$

Thus, allowing for credit card borrowing does not alter the mechanism.

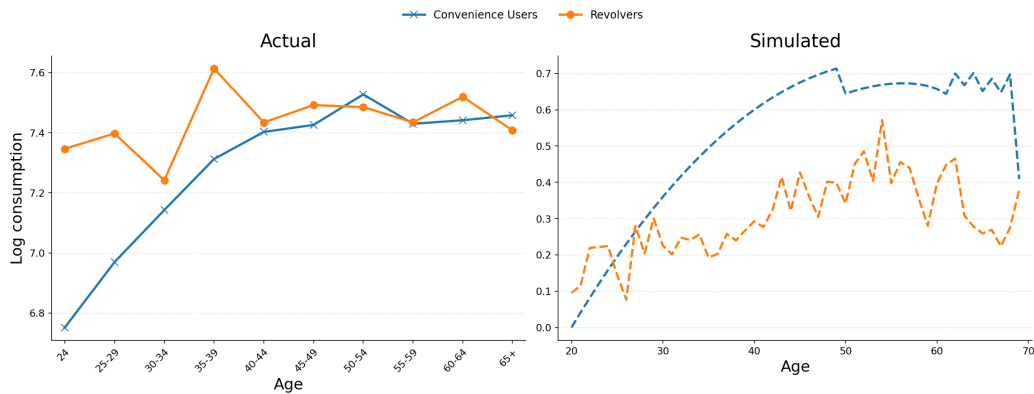
assets, revolving cardholders can maintain financial buffers for consumption smoothing, generating levels of financial comfort comparable to those of their non-revolving counterparts. Consequently, the model suggests that revolving debt should not be interpreted solely as a symptom of financial vulnerability, but rather as a potential cost of self-control.

Simulation: To illustrate the dynamic implications of the “investor-spenders” mechanism, I extend Eq.(5) to a finite-horizon setting and numerically solve the resulting model. The exercise is intended to illustrate that the framework can qualitatively reproduce the consumption and debt patterns observed in the data, rather than to estimate structural parameters or match empirical moments. Accordingly, the parameter values are chosen for illustration and are not calibrated to the data.

Generally, the simulated consumption dynamics aligns well observed pattern in the data. More specifically, *Revolvers* starts out at higher consumption level but experience less growth during prime working age, eventually falling below the consumption level of *Convenience users*. Thus, revolving cardholders exhibit a relatively flat consumption profile when compared to their non-revolving counterparts. Furthermore, simulated debt rises during early life-cycle stages before steadily declining, reflecting the dynamics observed in the data. Details regarding the simulation exercise are available in Appendix A.6

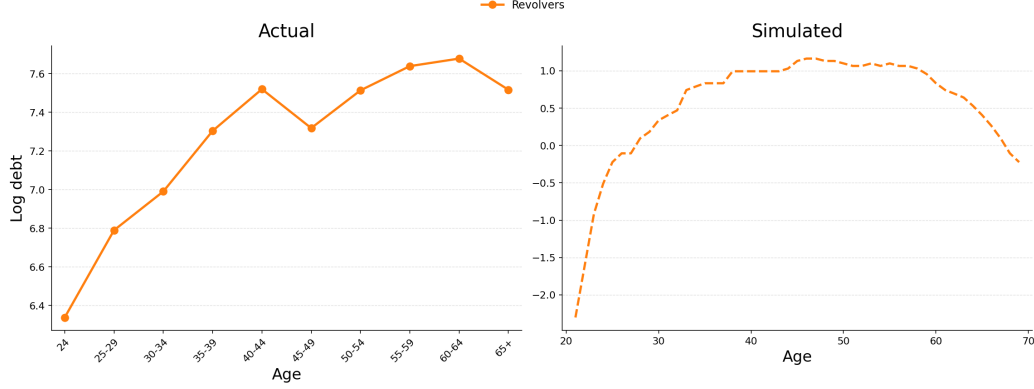
Notes: Simulated paths assume *Convenience users* exhibit time-consistent discount factors whereas *Revolvers* have time-inconsistent discount factors.

Figure 8: Actual vs. Simulated consumption path



Notes: Simulated paths assume *Convenience users* exhibit time-consistent discount factors whereas *Revolvers* have time-inconsistent discount factors.

Figure 9: Actual vs. Simulated outstanding balance



7 Conclusion

With the total value of credit card debt exceeding CAD 122 billion in Canada (Equifax Canada, 2024), regulatory concerns about consumer financial vulnerability are both understandable and warranted (Bilyk and Peterson, 2015; Xiao, 2024). Nevertheless, these concerns predicate on the perception that consumers only revolve on credit cards as last resorts to cover essential expenditures (CBC News, 2022; Fulford and Schuh, 2024). In this paper, I analyze differences between *Convenience users* and *Revolvers* and assess the validity of the narratives around credit card debt. Leveraging micro data, I find that the prevailing images of *Revolvers* as struggling consumers with poor financial well-being and no savings are largely unsubstantiated. In spite of lower income and education levels, *Revolvers* exhibit similar levels of assets ownership and only slight higher levels of liabilities in comparison to *Convenience users* (Figure 2). Moreover, both groups of cardholders report near identical levels of comfort in managing their finances (Figure 3).

To better understand the drivers of revolving behavior, I examine how key socioeconomic factors relate to both the likelihood of revolving and the amount of outstanding credit card debt. While *Revolvers* earn about CAD 10,000 less than *Convenience user* annually, income has rather negligible impacts on both the propensity to revolve as well as how much to borrow

conditional on revolving. Conditional on assets and liabilities, an CAD 10,000 increase in pretax income reduces the probability of revolving by only 1 pp while having no significant income on outstanding balance in the main specification.

By contrast, ownership of an investment account predicts a 10 pp lower probability of revolving, whereas being a mortgagor predicts a 7 pp higher probability. Because investment accounts and mortgages capture opposing liquidity conditions, these divergent effects provide suggestive evidence of liquidity-driven motives for revolving behavior.

Moving beyond demographic differences, this paper documents new evidence regarding consumption dynamics of *Revolvers* and *Convenience users*. Contrary to standard predictions (Fulford and Schuh, 2024), data suggest consumption levels of *Revolvers* remains relatively flat over time. This pattern is consistent with the idea that revolving cardholders use credit card debt as a tool to smooth out their consumption (Figure 6, Table 6).

To be clear, this paper does not argue that *Revolvers* never experience financial distress. Rather, it argues that revolving credit card debt is not, by itself, a reliable indicator of financial distress as that concept is typically understood.

A Appendix

A.1 Income definition

The DWPT asks participants to choose from 26 income brackets. First 20 brackets are equally spaced with a CAD 5000 step, starting from under CAD 5000 to CAD 95,000-CAD 99,999. The last 6 income brackets are: CAD 100,000- CAD 124,999, CAD 150,000-CAD 199,999 , CAD 200,000-CAD 249,999 and over CAD 250,000.

A.2 Additional Tables And Figures

Figure 10: 75% of *Revolvers* carry outstanding balance consistently

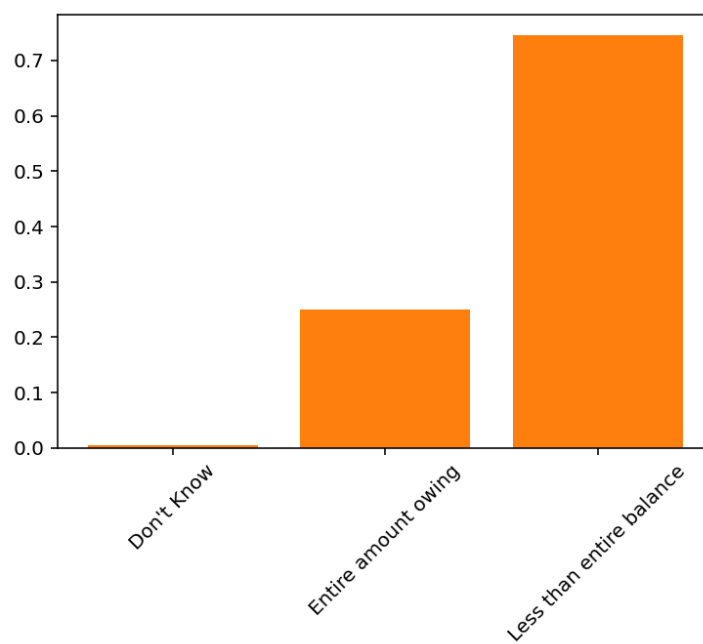


Figure 11: Elbow plot - Descriptive

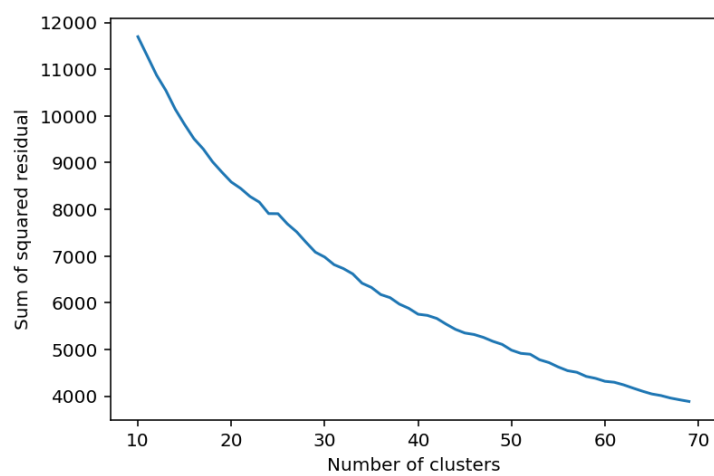
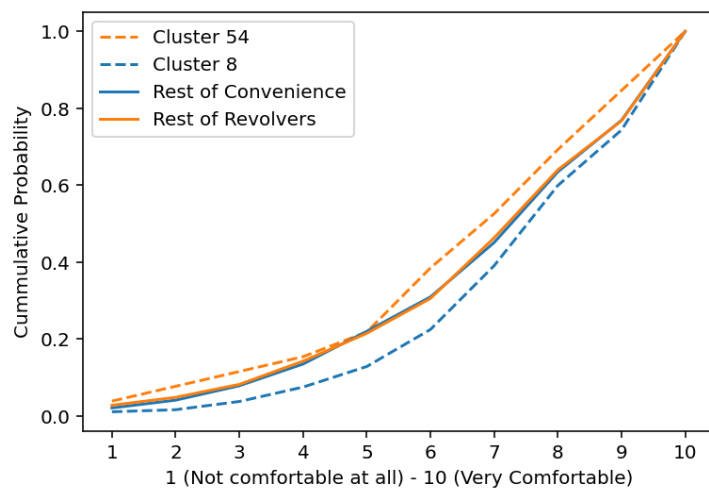


Figure 12: Financial comfort: Cluster 54 vs Rest of *Revolvers*



Cluster 8 and 54 consist solely of *Convenience users* and *Revolvers*, respectively. *Convenience users* and *Revolvers* exclude cardholders in cluster 8 and 54.

Figure 13: The survey asks: "How would you describe your current credit rating?"

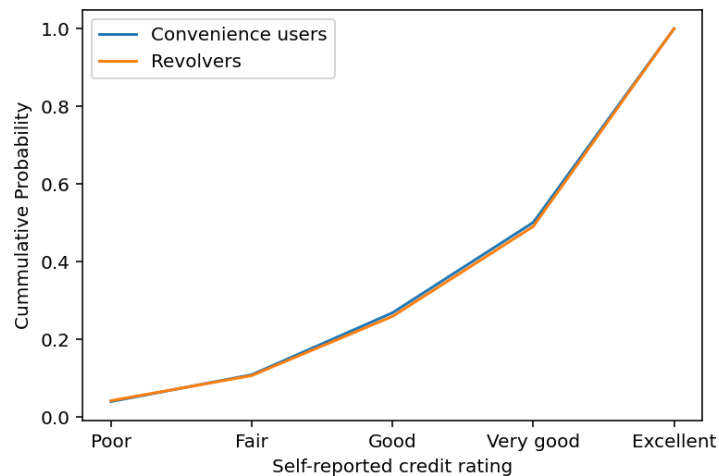


Table 7

Table a. Balance check: misreporting - *Revolvers* subsample

A *Revolver* is classified “Correctly report” if they choose other categories other than “No Debt” when asked about the amount of total financial liabilities.

Variable	Misreport	Correctly report	p-value
Pretax Income (CAD/year)	72481	77,172	0.130
Age	46.02	45.75	1.000
Household size (persons)	2.41	2.34	0.668
Number of credit cards	1.82	1.83	1.000
Amount of outstanding balance (CAD)	4,137	3,968	1.000
Share with Bachelor Degree (%)	36.4	35.0	1.000
Share with Mortgage (%)	36.1	42.0	0.064
Share with Investment Acc (%)	43.3	46.7	0.668

Table b. *Convenience users* vs *Revolver*-“No debt” subsample

Married Individuals includes both married couples and common-law partnerships

Variable	<i>Convenience users</i>	<i>Revolvers</i>
Pretax Income (CAD/year) (\$)	85,807	75,052
Age	51.65	45.31
Share with Bachelor Degree (%)	49.8	34.3
Share with Mortgage (%)	30.5	38.9
Share with Investment Acc (%)	70.0	48.1
Share of Male (%)	48.2	44.8
Share of Married Individuals (%)	63.0	61.7
Household size (persons)	2.12	2.10
Number of credit cards	1.82	1.83

Table 8: Balance check: Observed vs Missing- Participant that did not select “No debt”

Observations are classified “Missing” if they did not select “No debt”
and did not report a comfort level.

Variable	Observed	Missing	p-value
Age	50.56	48.79	0.0006
Pretax Income (CAD/year)	83,131	85,178	0.691
Number of credit cards	1.79	1.78	1.000
Household size (persons)	2.54	2.55	1.000
Share with Bachelor Degree (%)	45.4	48.0	0.533
Share with Mortgage (%)	33.1	33.0	1.000
Share with Investment Acc (%)	56.8	55.6	1.000
Outstanding Balance (CAD)	956	800	0.533
Share of <i>Revolvers</i> (%)	22.2	21.9	1.000

Table 9: Share of primary card

Primary card is defined as the most often used credit cards
Share is calculated as the share of spending/outstanding
balance on the primary card

Primary share (%)	Mean	Stdev	p25	p50	p75
Credit card spending	80.8	26.6	66.7	97.1	1
Outstanding balance	76.4	34.1	55.6	1	1

Table 10: Income Dynamics

	$\log(\text{Income})$
<i>Revolver</i>	-.121 (.156)
<i>Age</i>	.041*** (.003)
<i>Revolver</i> $\times$ <i>Age</i>	.004 (.007)
<i>Age</i> ²	-.0004*** (.00003)
<i>Revolver</i> $\times$ <i>Age</i> ²	-.00008 (.00007)
<i>University</i>	.335*** (.012)
<i>Children</i>	.011 (.010)
<i>Household Size</i>	-.002 (.007)
<i>Constant</i>	7.616*** (.071)
R^2	.098
Observations	12,056

Notes: The dependent variable is the logarithm of annual household income. Heteroskedasticity-robust HC0 standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

A.3 Double Machine Learning Estimate Of Financial Comfort

The empirical framework consists of two nuisance components:

$$\begin{aligned} \text{Comfort}_i &= g(\mathbf{1}_{\text{Rev},i}, \mathbf{Z}_i) + \nu_i, \\ \Pr(\mathbf{1}_{\text{Rev},i} = 1 \mid \mathbf{Z}_i) &= m(\mathbf{Z}_i) \end{aligned} \tag{6}$$

where $g(\cdot)$ captures the relationship between reported financial comfort, revolving status and observable characteristics. Meanwhile, $m(\mathbf{Z}_i)$ denotes the

propensity score, or the predicted probability that individual i is a *Revolver* conditional on observables.

Following the standard Double Machine Learning approach (Chernozhukov et al., 2018; Bach et al., 2022), I treat $g(\cdot)$ and $m(\cdot)$ as nuisance functions and estimate them using random forests.²¹ This approach allows for flexible, nonlinear relationships between financial comfort and observable characteristics without imposing a specific functional form. However, machine-learning estimators can introduce regularization bias because of model selection and hyperparameter tuning. To reduce this bias, I use doubly robust estimator which are less sensitive to small estimation errors in the nuisance functions.

The estimated observation-level orthogonal score (i.e., $\tilde{\theta}_{Rev}^i$) is computed as:

$$\hat{g}(1, \mathbf{Z}_i) - \hat{g}(0, \mathbf{Z}_i) + \frac{\mathbf{1}_{Rev,i} (\text{Comfort}_i - \hat{g}(1, \mathbf{Z}_i))}{\hat{m}(\mathbf{Z}_i)} - \frac{(1 - \mathbf{1}_{Rev,i}) (\text{Comfort}_i - \hat{g}(0, \mathbf{Z}_i))}{1 - \hat{m}(\mathbf{Z}_i)}.$$

Averaging these scores across individuals provides an estimate of the average conditional difference in financial comfort associated with being a *Revolver*.

Intuitively, $\tilde{\theta}_{Rev}^i$ compares individual i 's predicted comfort of a *Revolver* with that of an observationally equivalent *Convenience user* (i.e., $\hat{g}(1, \mathbf{X}_i) - \hat{g}(0, \mathbf{X}_i)$). The adjustment terms then correct for prediction errors in the outcome model ($\hat{g}(\cdot)$) using the observed comfort levels and estimated propensity scores, making the final estimate less sensitive to regularization bias in the nuisance functions. The model is cross-fitted to avoid overfitting bias (Bach et al., 2022).

Table 11 reports the Double Machine Learning estimate of the conditional relationship between revolving status and reported financial comfort. The average orthogonal score, $\mathbb{E}[\tilde{\theta}_{Rev}^i]$, is close to zero and statistically insignificant. This result suggests that *Revolvers* do not report systematically lower financial comfort than comparable *Convenience users*, even after flexibly controlling for observable financial conditions.

²¹Specifically, I use `sklearn`'s `RandomForestClassifier` to estimate the propensity score and `RandomForestRegressor` to estimate the outcome model.

Table 11: Double ML Estimates: Financial Comfort

	<i>Comfort</i>
$\mathbb{E}[\tilde{\theta}_{Rev}^i]$	-.020 (.060)
Observations	5,722
Controls	Yes
Outcome learner	Random Forest
Propensity learner	Random Forest

Notes: Standard errors are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4 Additional robustness check for determinant of debt

Given fixed ρ , intensive margin relationships are estimated as:

$$\log \text{Balance} = \beta_0 + \log X\beta_1 + D\beta_2 + Z_2\beta_3 + \lambda \frac{\phi(X, D, Z_2)}{\Phi(X, D, Z_2)}$$

While this equation can be estimated using a full-information maximum likelihood approach, such an implementation suffers from convergence issues as Z_2 contains multiple categorical control variables.

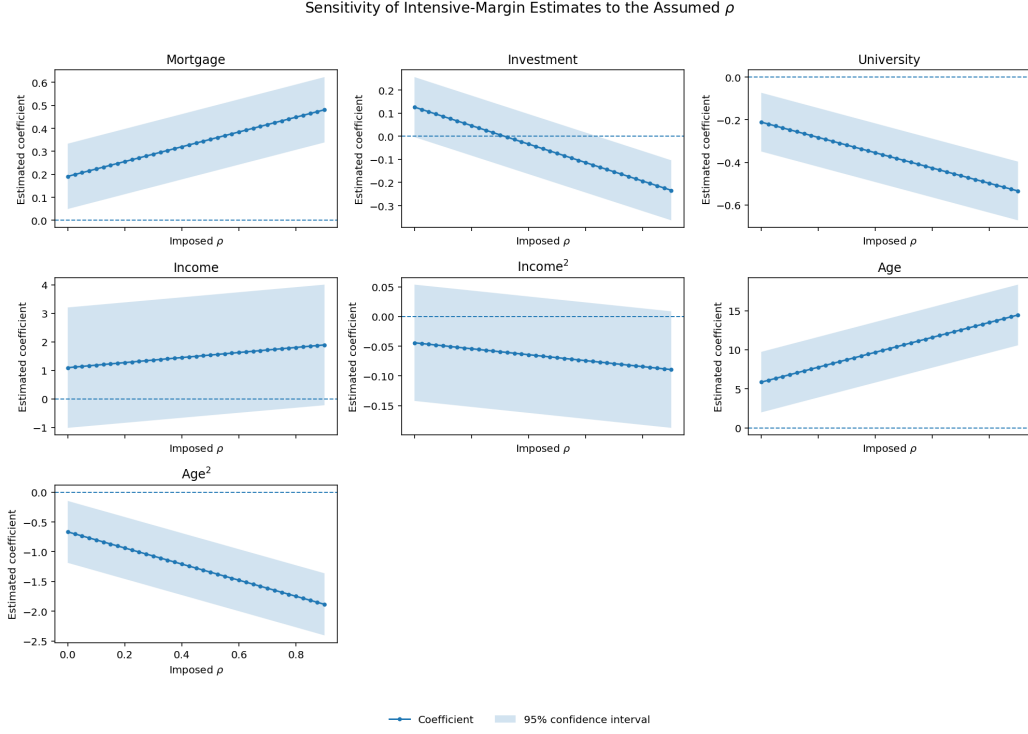
Let P_{X,D,Z_2} denote the projection matrix onto the plane spanned by X, D, Z_2 . Given ρ , $\hat{\sigma}_\nu$ can be estimated from using OLS:

$$(I - P_{X,D,Z_2}) \log \text{Balance} = \sigma_\nu (I - P_{X,D,Z_2}) \rho IMR$$

The intensive-margin relationships given ρ is thus,

$$\log \text{Balance} - \rho \hat{\sigma}_\nu IMR = \beta_0 + \log X\beta_1 + D\beta_2 + Z_2\beta_3$$

Figure 14: Sensitivity Analysis



A.5 Heterogeneity among *Convenience users* and *Revolvers*

So far in the analysis, we have treated *Convenience users* and *Revolvers* as internally uniform groups cardholders. While Section 3.1 has clearly indicated that concerns regarding financial vulnerabilities among *Revolvers* are overstated, such analysis obscures substantial heterogeneity across different subgroups of *Revolvers* and *Convenience users*. In order to unearth the underlying patterns in the data, I will conduct a cluster analysis to identify subgroups of credit cards users. More specifically, I will apply the k-means algorithm to assign users to different clusters based on a set of predefined characteristics. The goal is to create clusters whose members are homogeneous while maximizing heterogeneity between clusters. K-means achieves this by partitioning observations to minimize intra-cluster variance (the sum of squared distances between points and their cluster centroids). Since the

total variance in the dataset is fixed, reducing intra-cluster variance inherently increases inter-cluster variance, making clusters more distinct. In this context, the algorithm not only captures differences between *Revolvers* and *Convenience users* but also reveals heterogeneity within subgroups of each.

Clustering represents a valuable complement to regression analysis, especially when theoretical guidance is limited or ambiguous. Whereas regression aims to establish causal links between a set of regressors and an outcome variable, clustering approaches attempt to capture the inherent structure of the data. To estimate regression models, the analyst must predetermine the functional form of the relationship, including whether to include higher-order terms and/or interaction effects between independent variables. Often, these modeling choices require either a priori domain knowledge or theoretical justification to avoid potential bias due to misspecification. Alternatively, one can start with a fully saturated model (i.e., a model including all possible interaction terms) and remove the least significant predictors iteratively until arriving at a satisfactory specification. This stepwise approach, however, suffers from serious statistical drawbacks, namely the tendency to produce artificially narrow confidence intervals. Yet another approach is to apply non-parametric regression techniques, which avoid imposing any functional form restrictions. These methods come with their own challenges, including heavy computational burden in high-dimensional spaces (i.e., the curse of dimensionality) and difficulties in forming confidence intervals. Like nonparametric regressions, clustering algorithms do not impose functional restrictions on the relationships between variables. This flexibility accommodates varying relationships among the same variables across clusters, making clustering adept at identifying nonlinearities and structural breaks. In this sense, clustering provides a useful exploratory tool to inform researchers about the underlying data-generating process.

Implementation: To implement k-means, respondents are clustered based on six characteristics: (i) *Income*, (ii) *Age*, and a set of indicator functions reflecting whether they have: (iii) a mortgage, (iv) an investment account, (v) a bachelor’s degree and (vi) any revolving balance. These characteristics are selected specifically to test whether revolving consumers are more financially vulnerable. Because k-means uses Euclidean distance to measure similarity, the algorithm places more weight on variables with larger scale, resulting in unreliable clusters. To avoid scale-dependency, I standardize *Income* and *Age* to have $\mu_{Age} = \mu_{Income} = 0$ and $\sigma_{Age}^2 = \sigma_{Income}^2 = 1$. I determine the optimal number of clusters using the elbow method, which

balances minimizing intra-cluster variance with parsimony. The algorithm identifies a total of 60 clusters, of which 36 consist exclusively of either *Convenience users* or *Revolvers*, indicating meaningful differences between these groups across the selected dimensions. Table 12 summarizes results of the clustering. Due to space limitations, I report only 20 of the 60 clusters, chosen to illustrate how clustering can reveal nonlinearities in the data. The full summary statistics for all clusters are provided in Appendix A.5 (Table 12).

While Section 3.1 established that, on average, *Revolvers* are not in more precarious financial situations than their non-revolving counterparts, subsequent clustering analysis reveals that the common stereotype of financially struggling borrowers does apply to a subset—approximately 6%—of *Revolvers*. The subgroup in question (Cluster 54, Table 12) consists primarily of near-retirement individuals with lower income and educational attainment. With an average income around CAD 44,000, these *Revolvers* earn significantly less than the average revolving user, whose income is approximately CAD 75,000 (Table 2). They are also considerably older (60 vs 46 years) and are less likely to hold a university degree (18% vs 35.5%). Further analysis shows that this group also feels less comfortable with their finances (Figure 15, Appendix A.5) in comparison to *Convenience users* with similar demographic profiles (i.e., cluster 8) as well as other *Revolvers* (Figure 15, Appendix A.5), suggesting that they may indeed experience financial distress at a higher rate.

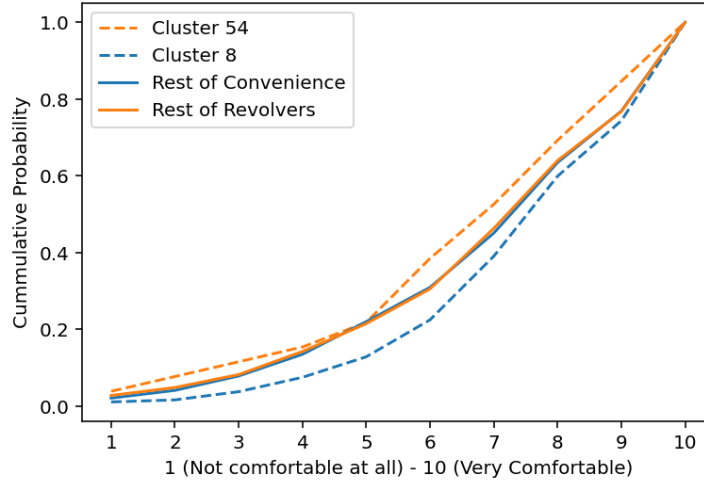
Observe that while the *Revolvers*' average income is lower than that of *Convenience users* (Table 2), our cluster analysis reveals distinct subgroups of *Convenience users* with income levels comparable to those of *Revolvers* (e.g., Cluster 2 vs. Cluster 48; Cluster 12 vs. Cluster 55). This pattern suggests a nonlinear association between household income and borrowing behaviors. A similar nonlinear relationship emerges when examining age effects on revolving status (e.g. Cluster 3, 6 vs Cluster 54, 59; Cluster 2,5 vs Cluster 48 ,56).

K-means also reveals an intriguing interactions between education level and age among *Revolvers*. Focusing on clusters composed entirely of *Revolvers* (i.e., Cluster 48, 49, 50, 51, 52, 54, 55, 56, 57, 58, 59 and 60), we observe significant disparities in education levels between clusters of predominantly older cardholders (e.g., Clusters 49, 54, and 59) and those made up mostly of younger cardholders (e.g., Clusters 50, 52, 55, and 60). Except for Cluster 48 and 57, younger clusters have higher share of cardholders with a bachelor's degree, whereas older clusters comprise mostly consumers with-

out a degree. This pattern indicates the effects of education on consumers' propensity to revolve are heterogeneous across age groups, especially around retirement age (65 and over)-a hypothesis that is confirmed by our estimates in Section 4.

Collectively, these patterns exemplify the value of clustering analysis in shaping subsequent modeling decisions. They provide empirical justification for incorporating quadratic (or higher-order) terms for income and age, along with education-age interaction terms to account for the heterogeneous effects of education across age groups.

Figure 15: Vulnerable Clusters



A.6 Simulation Details

Convenience users have time-consistent discount factor (β_I), these consumers do not have to rely on illiquid assets as commitment devices, and invest only in liquid assets. Hence, the Bellman equation is given by:

$$V^{Conv} = \max_{L, b_{t+1}} U(C_t) + \beta_I \mathbb{E} [V^{Conv}(y_{t+1}, L_{t+1})] \quad (7)$$

subject to

$$\begin{aligned} C_t + I_t &= y_t - (1 + r_{cc})b_t + (1 + r_l)L_t \\ L_{t+1} &= L_t + I_t \end{aligned}$$

Table 12: Summary Statistics Across Clusters

Name	Share with Mortgage (%)	Share with Investment Acc (%)	Share with Bachelor Degree (%)	Share of <i>Renoteers</i> (%)	Pretax Income (CAD/year)	Age	Observations
1	1.09	100.00	14.13	0.00	31,249.95	52.96	184
2	4.98	0.00	0.00	0.00	32,758.55	25.04	261
3	0.00	0.00	0.00	0.00	35,822.14	60.23	298
4	11.62	0.00	0.00	0.00	37,012.44	42.67	241
5	0.00	0.00	0.00	0.00	37,104.07	75.03	221
6	1.30	0.00	100.00	0.00	38,051.90	27.75	231
7	5.43	0.00	100.00	0.00	39,844.93	47.26	129
8	0.00	100.00	0.00	0.00	44,192.04	64.59	427
9	0.00	100.00	0.00	0.00	67,337.66	39.54	154
10	100.00	100.00	0.00	0.00	73,266.93	44.37	251
11	100.00	0.00	85.11	0.00	80,718.09	33.54	188
12	100.00	0.00	0.00	0.00	84,185.52	50.75	221
13	0.00	0.00	100.00	0.00	87,722.77	31.53	202
14	8.86	0.00	0.00	0.00	89,430.38	25.82	158
15	100.00	100.00	100.00	0.00	89,895.83	33.52	288
16	0.00	100.00	100.00	0.00	90,270.83	31.34	240
17	100.00	0.00	64.49	0.00	137,616.82	36.96	107
18	100.00	100.00	100.00	0.00	141,293.10	42.33	290
19	100.00	53.06	0.00	0.01	46,147.96	67.23	196
20	0.38	100.00	0.38	0.01	43,943.40	76.24	265
21	100.00	100.00	100.00	0.02	58,496.24	46.77	133
22	0.30	100.00	0.00	0.03	87,247.02	71.63	336
23	11.27	90.14	100.00	0.03	140,492.96	68.44	213
24	4.73	96.15	100.00	0.04	89,319.53	72.74	338
25	2.60	0.00	0.00	0.04	81,614.58	67.93	192
26	2.67	100.00	100.00	0.04	42,844.43	30.60	225
27	100.00	100.00	0.00	0.05	93,027.03	61.29	185
28	0.00	92.42	100.00	0.05	139,646.46	52.98	198
29	11.15	100.00	100.00	0.05	46,147.54	69.16	305
30	16.33	84.35	72.11	0.05	200,000.00	61.43	147
31	54.17	100.00	0.00	0.06	97,638.89	26.26	144
32	0.00	0.00	87.06	0.06	138,823.53	32.64	85
33	0.00	100.00	0.00	0.07	93,661.29	56.86	310
34	0.00	100.00	100.00	0.07	84,488.19	54.35	381
35	100.00	100.00	100.00	0.09	97,971.01	56.05	276
36	97.81	97.27	0.00	0.09	140,027.32	48.04	183
37	92.04	85.84	84.96	0.09	250,000.00	45.44	113
38	100.00	0.00	100.00	0.09	95,688.41	55.90	138
39	14.56	80.58	79.61	0.10	250,000.00	61.00	103
40	8.77	89.47	0.00	0.12	143,128.66	63.04	171
41	0.00	0.00	100.00	0.13	94,689.12	59.25	193
42	0.71	100.00	80.14	0.13	141,489.36	31.96	141
43	10.58	0.00	100.00	0.15	42,460.32	69.34	189
44	0.00	0.00	0.00	0.16	103,480.66	53.20	181
45	85.58	80.00	73.49	0.19	200,000.00	42.47	215
46	5.94	50.50	68.32	0.23	225,742.57	27.99	101
47	11.59	100.00	0.00	0.34	36,189.01	26.41	164
48	8.42	0.00	0.00	1.00	36,816.00	27.69	190
49	0.00	0.00	0.00	1.00	40,692.00	66.61	159
50	10.40	0.00	100.00	1.00	46,936.00	34.10	173
51	2.07	0.00	7.88	1.00	41,183.00	48.81	241
52	0.00	100.00	48.68	1.00	68,730.00	35.23	189
53	100.00	100.00	18.75	1.00	54,861.00	57.69	144
54	0.00	100.00	18.47	1.00	44,363.00	59.62	157
55	100.00	67.32	100.00	1.00	86,667.00	37.05	153
56	2.63	0.66	35.53	1.00	97,171.00	31.78	132
57	100.00	0.00	0.00	1.00	89,053.00	41.63	132
58	0.00	100.00	48.68	1.00	68,730.00	35.23	189
59	0.00	0.00	0.00	1.00	40,692.00	66.61	159
60	9.34	62.77	81.02	1.00	141,241.00	42.77	137

where L_t, b_t denote the stock of liquid assets and outstanding balances, while r_l, r_c represent the corresponding interest rates, respectively.

By contrast, *Revolvers* exhibit time-inconsistent discount factor. Because the “spender” exhausts the stock of liquid assets and sell off the maximum allowed share of liquid assets, the “investor” would invest only in illiquid assets to maintain long-term savings. The within-period decisions are sequential: the investor first chooses illiquid investment I_t , after which the spender chooses credit card borrowing b_{t+1} . Let the stock of illiquid assets available to the spender after the investment decision be

$$A_t = (1 + r_k)K_t + I_t$$

Conditional on A_t , the spender solves

$$V_{Spend}^{Rev}(y_t, b_t, K_t; I_t) = \max_{b_{t+1}} U(C_t) + \beta_S \mathbb{E} [V_{Invest}^{Rev}(y_{t+1}, b_{t+1}, K_{t+1})] \quad (8)$$

subject to

$$\begin{aligned} C_t + I_t &= y_t - (1 + r_{cc})b_t + p\bar{\theta}K A_t + b_t + 1, \\ K_{t+1} &= (1 - \bar{\theta}K)A_t, \\ 0 &\leq b_{t+1} \leq \bar{D}_t = \kappa y_t. \end{aligned}$$

where r_k denotes the return on illiquid assets, and $p < 1$ is the liquidation value of the illiquid asset.

Let $b_{t+1}(y_t, b_t, K_t)$ denote the “spender” borrowing policy function. Anticipating this response, the investor solves

$$V_{Invest}^{Rev}(y_t, b_t, K_t) = \max_{I_t} U(C_t) + \beta_I \mathbb{E} [V_{Invest}^{Rev}(y_{t+1}, b_{t+1}, K_{t+1})] \quad (9)$$

This formulation makes explicit that the investor values future resources more strongly than the spender and chooses illiquid investment while anticipating the spender’s subsequent borrowing decision.

$$V_{Invest}^{Rev}(y_t, b_t, K_t) = \max_{I_t} U(C_t) + \beta_I \mathbb{E} [V_{Invest}^{Rev}(y_{t+1}, b_{t+1}, K_{t+1})] \quad (10)$$

subject to

$$\begin{aligned} C_t + I_t &= y_t - (1 + r_{cc})b_t - p\bar{\theta}_k A_t + b_{t+1} \\ K_{t+1} &= (1 - \bar{\theta}_K)(1 + r_k)(K_t + I_t) \\ A_t &= (1 + r_k)K_t + I_t \end{aligned}$$

where I_t, r_k denote investment in and interest rate of illiquid assets. Finally, A_t denotes the stock of illiquid assets available for liquidation by the “spender”.

Credit car borrowing is determined by:

$$V_{Spend}^{Rev}(y_t, b_t, K_t) = \max_{b_{t+1}} U(C_t) + \beta_S \mathbb{E} [V_{Invest}^{Rev}(y_{t+1}, b_{t+1}, K_{t+1})] \quad (11)$$

subject to the same constraints and a borrowing limit

$$b_{t+1} \leq \bar{D}_t = \kappa y_t$$

The income path is deterministic according to the following process:

$$y_t = w_t Y_t^W + (1 - w_t) Y_t^R$$

where

$$\begin{aligned} Experience_t &= \min\{t, 65\} - 20 \\ w_t &= \frac{1}{1 + \exp \frac{t-65}{1.5}} \\ Y_t^W &= 1 + 0.062 Experience_t - 0.000886 Experience_t^2 \\ Y_t^R &= 0.75 Y_{65}^W \end{aligned}$$

Incomes are normalized so that income at the initial age equals one.

A.7 Robustness Checks With Alternative Definition Of Revolving

app:alternative-definition This section reproduces Table 2 and Figures 2 and 3 using an alternative definition of *Revolvers* based directly on respondents’ usual repayment behavior. Under this definition, *Revolvers* are respondents who do not select “Entire amount owing” when asked whether they usually pay the entire amount owing, the minimum amount owing, or an amount somewhere in between. The results presented here are consistent with the main results.

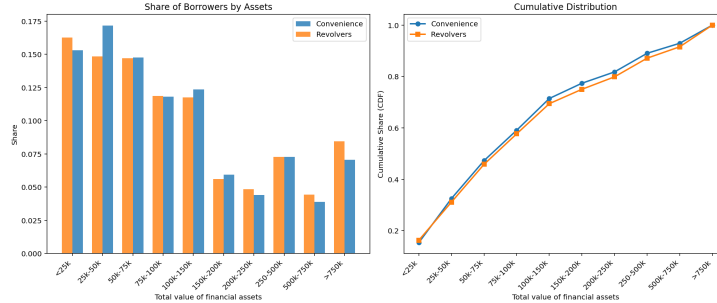
Table 13: Simulation Parameters

Parameter	Description	Time-consistent	Investor-spender
<i>Life-cycle income profile</i>			
T	Number of periods	50	50
Starting age	Age in the first period	20	20
Retirement age	Beginning of retirement	65	65
<i>Preference and return parameters</i>			
β_I	Investor's discount factor	0.97	0.97
β_S	Spender's discount factor	—	0.78
γ	Coefficient of relative risk aversion	2.00	2.20
R_L	Gross return on liquid assets	1.03	—
R_K	Gross return on illiquid assets	—	1.07
R_{cc}	Gross credit-card interest rate	1.20	1.20
$\bar{\theta}_K$	Maximum share of illiquid assets liquidated	—	0.12
p	Liquidation value of illiquid assets	—	0.97
κ	Borrowing limit as a share of income	—	0.70

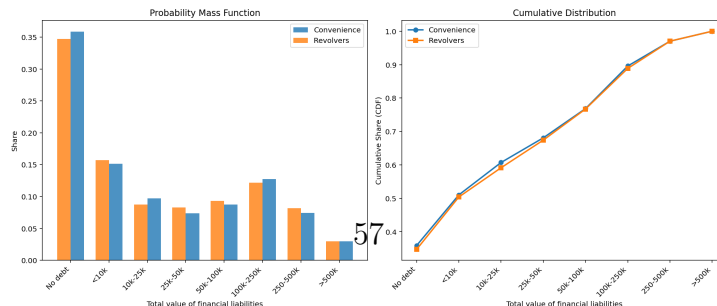
Notes: Both models use the same hump-shaped life-cycle income profile. The time-consistent benchmark permits saving in liquid assets, whereas the investor-spender model permits investment in illiquid assets and distinguishes between the investor's and spender's discount factors.

The parameter values are selected for illustrative purposes and are not calibrated to match empirical moments.

Figure 16: Cumulative distribution of financial assets and liabilities



(a) The DWPT asks: “*What is the total value of your household’s financial assets?*”



(b) The DWPT asks: “*Which of the following categories best describes your household’s TOTAL FINANCIAL LIABILITIES?*”

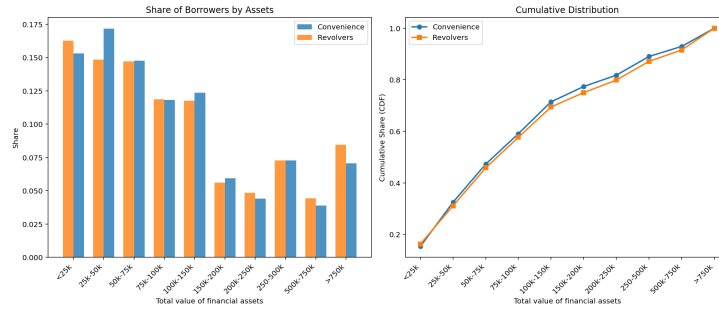
Table 14: Demographics of *Convenience users* and *Revolvers*

Variable	<i>Convenience users</i>	<i>Revolvers</i>
Pretax Income (CAD/year) (\$)	87,736	72,203
Age	51	46
Share with Bachelor Degree (%)	50	33
Share with Mortgage (%)	31.1	39.89
Share with Investment Acc (%)	64.7	43.2
Share of Male (%)	50	46
Share of Married Individuals (%)	62.5	60.7
Household size (persons)	2.4	2.38
Number of credit cards	1.8	1.79

Married Individuals includes both married couples and common law partnership.

Revolvers are cardholders who usually do not repay the whole balance.

Figure 17: The DWPT asks: “How comfortable are you managing your current debt load?”



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